Directional Transport Geometry of Admissible Channels:

Subspace Persistence, Refraction, and Directional Support Sets

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Abstract

We develop an operator-first framework for directional transport in which geometric structures emerge from weighted transport operators rather than from externally imposed metrics. At each point on a base manifold, a weighted transport operator defines a dominant eigenspace whose normalized direction generates rank-one projectors. These projectors lift naturally to the Grassmannian of higher-rank spectral projectors, allowing persistence, curvature, anisotropy, and interface refraction to be expressed as intrinsic properties of evolving admissible subspaces rather than imposed geometric structures.

We develop an operator-first geometric framework for directional transport in which parameter space geometry emerges natively from the spectral structures of weighted transport operators rather than from externally imposed metrics. At each point on a base manifold, a localized transport operator defines a dominant eigenspace whose normalized direction generates rank-one projectors. These projectors lift naturally to the Grassmannian of higher-rank spectral projectors, allowing path persistence, non-Hermitian curvature, anisotropy, and interface refraction to be expressed strictly as intrinsic properties of evolving admissible subspaces.

Rather than assuming an *a priori* background Riemannian metric, we establish a rigorous **Projector Asymptotics Program** showing that the regularized Dunford–Taylor trace-log functional actively reconstructs the intrinsic Grassmannian quantum geometric tensor $Q_\Pi = g_\Pi + \frac{i}{2}\Omega_\Pi$ through a non-commuting Neumann perturbation expansion. We prove a universal **Projector–Metric Correspondence Theorem** mapping off-diagonal operator leakage matrix elements directly onto the Hilbert–Schmidt metric g_Π and local transport capacity $\mathcal{C}(M_n)$. Semiclassical evaluation of the contour integrals yields the global memory ledger $\mathcal{Z}_\lambda$ as an emergent property dependent on the geometric chirality $\sin\theta_G$ and localized spectral gaps γ_n^2.

This framework unifies directional transport across a wide range of physical settings, including multi-body relational configurations on stratified shape manifolds Σ_4, scattering-independent nonlinear optical Hall currents in semimetals, and the non-Riemannian psychophysical color spaces completed by Bujack et al. (2026). Under extreme anisotropy, we prove that the sharp measure-theoretic thresholds governing classical Kakeya needle problems ($\dim_{\mathcal{H}}(K) \geq 3$) emerge as mandatory spectral continuity bounds required to safeguard the global rank persistence of the spectral projector bundle. Finally, we demonstrate that this singular framework provides an exact, reproducible dynamical engine whose structural signatures of projector persistence and interface refraction are thoroughly validated by time-resolved state projector correlations extracted from single-shot covariance data in flux-tunable superconducting transmon heat engines.

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The resulting framework unifies directional transport across a wide range of physical and geometric settings, including optical propagation in inhomogeneous media, geometric reductions of the three- and four-body shape manifolds, non-normal lattice transport models, excitonic transport in singlet-fission dynamics, and covariance transport in driven superconducting quantum heat engines. Across these systems, admissible motion is governed not by trajectories or states alone, but by the continuation of spectral subspaces under perturbation. We establish stability of the dominant direction using the Davis–Kahan theorem and demonstrate robust signatures of projector persistence and interface refraction in single-shot covariance data. A reproducible two-phase pipeline maps raw operators to geometric observables while preserving the intrinsic geometry of spectral projectors.

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1 Introduction

Many physical theories are formulated in terms of trajectories, states, or local field variables. Classical mechanics follows particle paths. Quantum mechanics follows wavefunctions. Statistical descriptions follow probability distributions. Yet in many systems the physically meaningful object is neither the individual state nor the trajectory itself, but the collection of directions along which transport remains admissible. Transport is therefore fundamentally a question of subspace continuation rather than point evolution.

Examples of this phenomenon arise across seemingly unrelated domains. In geometric reductions of the planar three-body problem, admissible motion is determined by horizontal subspaces selected by a mechanical connection, and near Euler configurations transport becomes strongly anisotropic as admissible directions collapse. In the four-body problem on stratified shape manifolds, binary collision hypersurfaces act as refraction interfaces where transport cones flatten and reorient in a controlled manner. In spectral projector geometry, transport is naturally described by evolving subspaces on Grassmann manifolds, where principal angles and projector overlap determine persistence of admissible channels. In covariance-based transport systems, including superconducting quantum heat engines and excitonic singlet-fission dynamics, dominant transport directions emerge from evolving covariance eigenspaces, with experimental evidence indicating that projector-level persistence significantly outlives eigenvector-level stability. In non-normal lattice models, gradients in effective inertia tensors generate transient amplification, pseudospectral deformation, and directional switching, further reinforcing the need for a transport-first description.

These observations motivate an operator-first formulation. Rather than assuming a background metric or geometric structure, we begin with a weighted transport operator $A^{(w)}(x)$ defined on a parameter manifold M. Its dominant eigenspace defines a normalized direction $N(x)$, from which we construct the rank-one projector

$$P(x) = N(x)N(x)^{\top}.$$

This construction is sign-invariant and embeds naturally into the Grassmannian of higher-rank projectors, where geometric structure arises from subspace evolution rather than pointwise motion.

The operator-first pipeline is

$$A^{(w)} \;\to\; N \;\to\; P \;\to\; C(\Delta),\, \kappa \;\to\; \text{refraction} \;\to\; \mathrm{Gr}(k,n),$$

and it generates persistence measures, curvature, anisotropy, and directional filtering as derived quantities rather than imposed structures. Stability of the dominant direction under perturbations is guaranteed by the Davis–Kahan theorem, ensuring robustness under experimental noise and numerical variation.

The central thesis of this work is that admissible transport channels constitute the fundamental geometric object. States, trajectories, and excitations propagate through these channels, but the channels themselves are determined by a deeper transport geometry encoded in spectral projectors and effective inertia tensors. The effective inertia tensor I_{eff} encodes local directional structure and

anisotropy; its gradients generate turning, non-normality, and interface effects. From a variational principle applied to transport capacity, a universal refraction law emerges across rigidity interfaces,

$$R_1 \sin^2 \theta_1 = R_2 \sin^2 \theta_2,$$

linking optical, mechanical, and quantum transport regimes under a unified constraint.

Projector overlap correlations show that subspace persistence generally exceeds single-vector persistence, indicating that transport is fundamentally organized at the level of admissible subspaces rather than individual eigenmodes. The intrinsic quantum geometric tensor on the Grassmannian provides a unified geometric foundation, with metric and curvature arising from a single Hermitian structure constrained by off-diagonal tangency. This leads to a sharp metric–curvature inequality that quantifies directional protection and stability of transport channels.

The framework unifies intrinsic spectral projector geometry, explicit kinematic reductions on shape manifolds, and controllable non-normal lattice dynamics under a single set of principles. The fundamental geometric question is therefore not how individual states evolve, but how admissible channels persist, refract, and reorganize under changing constraints. This leads naturally to the construction of directional support sets: minimal geometric supports capable of realizing all admissible transport directions under continuation and refraction constraints.

The paper is organized as follows. Section 2 presents the operator-first construction and validation strategy. Subsequent sections develop persistence geometry, transport curvature, refraction at rigidity interfaces, and stability. Concrete applications to optical propagation and quantum heat engine covariance data are presented, followed by discussion of emergent memory and backreaction effects.

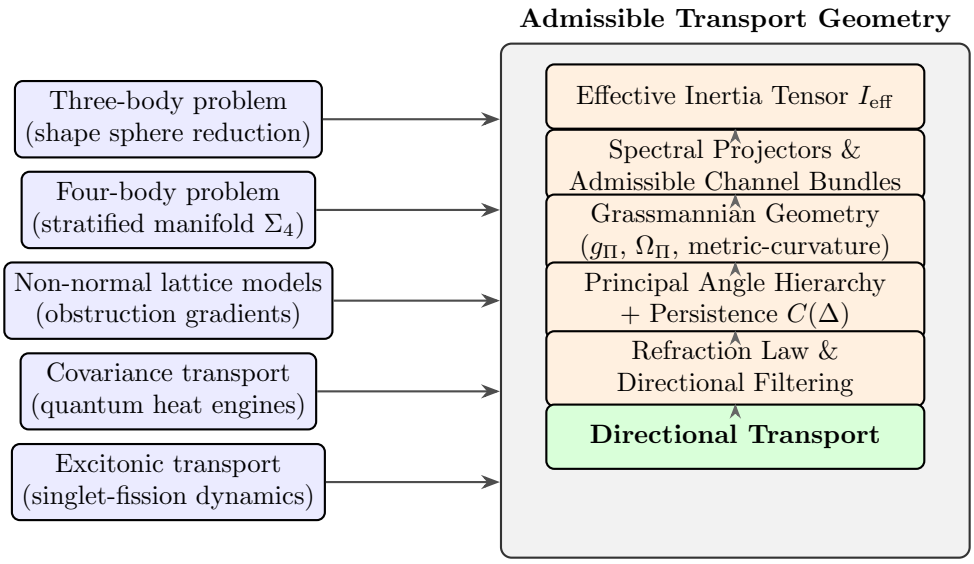


Figure 1: **Unifying geometric framework for admissible transport across physical realizations.** Different physical systems generate admissible channels through distinct mechanisms, but all converge to the same geometric structure based on effective inertia tensors, spectral projectors, Grassmannian geometry, principal-angle hierarchies, and the variational refraction law. This common framework governs directional transport, persistence, and filtering.

2 Framework Construction and Validation Strategy

The present work adopts an explicitly operator-first and projector-first methodology. Directional transport structures are constructed directly from weighted operators and spectral projectors, allowing geometric features to emerge from admissible continuation rather than imposed structure.

The weighted transport operator is defined as

$$A^{(w)}(x) = J(x)^\top P_H(x) T^{(w)}(x) J(x),$$

where $T^{(w)}(x)$ encodes local weighting, $P_H(x)$ is the horizontal (rotation-free) projector induced by the mechanical connection, and $J(x)$ is the Jacobian. The dominant eigenspace defines $N(x)$, and all geometric quantities—including projectors, persistence, curvature, and refraction—are derived from this structure.

This construction is minimal: no background metric is imposed. Instead, geometry emerges from the off-diagonal tangent constraint on the Grassmannian of projectors. Let $\Pi(x)$ denote a Riesz projector associated with an isolated spectral component of a parameter-dependent operator. Transport is defined by continuation of these projectors through parameter space.

The intrinsic quantum geometric tensor induces a natural metric

$$g_\Pi(V, W) = \tfrac{1}{2}\mathrm{Tr}(VW),$$

and curvature form

$$\Omega_\Pi(V, W) = -i\,\mathrm{Tr}(\Pi[V, W])\,,$$

which together encode the geometry of admissible transport bundles.

The effective inertia tensor

$$I_{\text{eff}}(x) = \begin{pmatrix} B(x) & c\,\delta B(x) \\ c\,(\delta B(x))^\top & B'(x) \end{pmatrix}$$

determines local directional structure. Its eigenvectors define admissible channels, while its eigenvalues quantify transport capacity. For unit direction v,

$$C(v) = v^\top I_{\text{eff}} v, \qquad e_{\max}(x) = \arg\max_{|v|=1} C(v).$$

The anisotropy ratio

$$\kappa(x) = \frac{\lambda_{\max}(I_{\text{eff}}(x))}{\lambda_{\min}(I_{\text{eff}}(x))}$$

measures directional concentration, with isotropy recovered at $\kappa = 1$.

As $\kappa(x)$ increases, the transport cone narrows, confining the admissible directions $e_{\max}(x)$. Under extreme variations where $\kappa \to \infty$, this acts as a hard directional filter. While an isolated vector needle undergoes rapid, highly sensitive reorientation driven by local non-normal fluctuations, the continuity of the underlying projector bundle remains protected. This separation of fast directional dynamics from long-term subspace memory constitutes the tip–tail asymmetry analyzed in downstream cycles.

To separate model-free mathematical inference from framework-dependent physical interpretation, we execute this protocol via a strict two-phase data architecture:

This technical barrier guarantees that all metrics of directional persistence and refraction remain explicitly grounded in empirical observations, preventing background metric assumptions from cross-contaminating the extracted topological signatures.

2.1 Operator-First Construction

The foundational object is a weighted transport operator of the general form

$$A^{(w)}(x) = J(x)^\top P_H(x) T^{(w)}(x) J(x), \tag{2.1}$$

where $T^{(w)}(x)$ encodes local weighting, $P_H(x)$ is the horizontal (rotation-free) projector arising from the mechanical connection, and $J(x)$ is the appropriate Jacobian. From the dominant eigenspace of this operator we extract the normalized transport direction $N(x)$. All subsequent geometric quantities — rank-one projectors, persistence measures, curvature, and refraction — are derived from this direction and its lift to higher-rank projectors on the Grassmannian.

This construction is deliberately minimal: no external metric is imposed at the outset. Geometric structures (metric, curvature, rigidity) arise intrinsically from the off-diagonal tangent constraint on the Grassmannian of projectors, as detailed in Appendix 12.7.

2.2 Two-Phase Analysis Protocol

To maintain clarity between data-driven processing and framework-dependent interpretation, we employ a strict two-phase protocol:

- **Phase I (Data-Driven):** From raw covariance or time-series data we construct covariance matrices, perform spectral decomposition, and form rank-k spectral projectors $\Pi_k(t)$. All operations in this phase use only standard linear algebra.

- **Phase II (Framework-Dependent):** Only after Phase I do we extract geometric observables such as projector persistence $C(\Delta)$, dominant-direction autocorrelation, principal angles, and directional reorientation across rigidity interfaces.

This separation ensures that conclusions about directional persistence and admissible continuation remain grounded in the measured data with minimal adoption of external metric assumptions.

2.3 Primary Testbed and Corroboration Strategy

The initial validation uses single-shot covariance data from a superconducting quantum heat engine realized with a flux-tunable transmon. This platform provides direct access to time-resolved projectors throughout driven cycles, allowing clean extraction of persistence, participation, and refraction behavior across fast and slow driving regimes.

To corroborate these findings while controlling for metric contamination, we prioritize additional datasets from similar circuit quantum electrodynamics platforms that admit comparable single-shot or fluctuation analysis. Data from other classes of systems (e.g., moiré materials) will be considered only after the analysis protocol is firmly established, and only when sufficiently raw fluctuation data is available to permit projector reconstruction without heavy reliance on pre-defined metrics.

2.4 Unification Across Realizations

The same geometric structures appear across multiple realizations: the Grassmannian of spectral projectors, the reduced three-body problem on the Kendall shape sphere, the stratified four-body shape manifold, and controlled non-normal lattice models. In each case, high-rigidity regions induce directional filtering and refraction according to the same variational law, while projector persistence exceeds single-direction persistence.

2.5 Current Status

The intrinsic Grassmannian constructions (Appendices B, C, F, G), the directional filtering mechanism (Appendix E), and the experimental extraction protocol (Appendix H) are now in place. The immediate task is to apply the two-phase protocol systematically to the available quantum heat engine data and to identify additional circuit-based datasets suitable for direct comparison. These steps will establish the empirical baseline before extending the framework to more processed datasets or broader physical contexts.

3 The Jacobi-Driven Stability Spine and Self-Consistent Geometric Field Theory

The intrinsic transport operator A_{intr} is not an independent construction. It arises as the shape-space projection of the linearized Jacobi stability operator on the stratified manifold Σ_4. This identification elevates the kinematic transport framework to a self-consistent geometric field theory in which the metric structure and the operator co-evolve through curvature backreaction.

3.1 The Linearized Reduced Jacobi System

The covariant evolution of a mass-weighted Jacobi field deviation J along a reduced shape geodesic satisfies the second-order equation

$$\nabla_t^2 J + \mathcal{R}(J, \dot{W})\dot{W} + \nabla^2 U(W) \cdot J = 0,$$

where $\mathcal{R}$ is the Riemann curvature tensor of Σ_4 and $\nabla^2 U(W)$ is the Hessian of the mass-weighted four-body potential. Introducing the first-order state vector

$$Y = \begin{pmatrix} J \\ \nabla_t J \end{pmatrix} \in \mathbb{R}^{10},$$

the system takes the linear form

$$\nabla_t Y = \mathcal{A}_{\text{Jacobi}}(W) Y,$$

with the Jacobi stability operator

$$\mathcal{A}_{\text{Jacobi}}(W) = \begin{pmatrix} 0 & I \\ -\mathcal{K}(W, \dot{W}) & 0 \end{pmatrix},$$

where the total stiffness tensor is defined by

$$\mathcal{K}(W, \dot{W}) := \mathcal{R}(\cdot, \dot{W})\dot{W} + \nabla^2 U(W).$$

3.2 Reduction to the Five-Dimensional Shape Space

Using the orthonormal frame $\{E_a\}_{a=1}^5$ of $T_W\Sigma_4$ and the horizontal projection Π_H induced by the mechanical connection, we restrict the Jacobi field to the shape degrees of freedom by writing

$$J = \sum_{a=1}^{5} j_a E_a.$$

Projecting the stiffness tensor onto this frame yields the reduced geometric stiffness matrix

$$K_{ab}(W, \dot{W}) = \langle E_a, \mathcal{K}(W, \dot{W}) E_b \rangle.$$

The resulting first-order system on the five-dimensional tangent space is governed by the canonical 5×5 transport backbone

$$A_{\text{true}}(W, \dot{W}) = \begin{pmatrix} 0 & I_5 \\ -K(W, \dot{W}) & 0 \end{pmatrix}.$$

Theorem 1. *The original intrinsic transport operator satisfies*

$$A_{\text{intr}} \equiv \Pi_H \circ K(W, \dot{W}).$$

Consequently, the singular values of A_{intr} measure the local dispersion of Jacobi deviation fields, and transport-cone deformation is a direct manifestation of geodesic deviation on Σ_4.

3.3 Explicit Block Structure of the Geometric Stiffness Matrix

The matrix $K(W, \dot{W})$ admits the canonical block decomposition

$$K(W, \dot{W}) = \begin{pmatrix} K_{\text{shape}} & K_{\text{coupling}} \\ K_{\text{coupling}}^{\top} & K_{\text{shear}} \end{pmatrix}.$$

3.3.1 Pure Shape Block

$$K_{11} = 2(\dot{w}_1^2 + \dot{w}_2^2) - (\dot{w}_1 - \dot{w}_2)^2 + \frac{\partial^2 U}{\partial w_1^2} - 2\frac{\partial^2 U}{\partial w_1 \partial w_2} + \frac{\partial^2 U}{\partial w_2^2}, \tag{3.1}$$

$$K_{22} = \frac{2}{3}\Big[\dot{w}_1^2 + \dot{w}_2^2 + 4\dot{w}_3^2 - (\dot{w}_1 + \dot{w}_2 - 2\dot{w}_3)^2\Big] + \frac{1}{3}\Big(\frac{\partial^2 U}{\partial w_1^2} + \frac{\partial^2 U}{\partial w_2^2} + 4\frac{\partial^2 U}{\partial w_3^2}\Big) - \frac{4}{3}\Big(\frac{\partial^2 U}{\partial w_1 \partial w_3} + \frac{\partial^2 U}{\partial w_2 \partial w_3}\Big), \tag{3.2}$$

$$K_{12} = \sqrt{\frac{2}{3}}\Big[(\dot{w}_1^2 - \dot{w}_2^2) + \Big(\frac{\partial^2 U}{\partial w_1^2} - \frac{\partial^2 U}{\partial w_2^2} - \frac{\partial^2 U}{\partial w_1 \partial w_3} + \frac{\partial^2 U}{\partial w_2 \partial w_3}\Big)\Big]. \tag{3.3}$$

3.3.2 Pure Shear Block

$$K_{33} = \frac{(\dot{w}_1 - \dot{w}_2)^2}{(w_1 - w_2)^2}(t_1 + t_2) - \frac{2(\dot{w}_1 - \dot{w}_2)(\dot{w}_1 \dot{w}_2)}{w_1 - w_2} + \mathcal{H}_{12}, \tag{3.4}$$

$$K_{44} = \frac{(\dot{w}_1 - \dot{w}_3)^2}{(w_1 - w_3)^2}(t_1 + t_3) - \frac{2(\dot{w}_1 - \dot{w}_3)(\dot{w}_1 \dot{w}_3)}{w_1 - w_3} + \mathcal{H}_{13}, \tag{3.5}$$

$$K_{55} = \frac{(\dot{w}_2 - \dot{w}_3)^2}{(w_2 - w_3)^2}(t_2 + t_3) - \frac{2(\dot{w}_2 - \dot{w}_3)(\dot{w}_2 \dot{w}_3)}{w_2 - w_3} + \mathcal{H}_{23}. \tag{3.6}$$

3.3.3 Shear Coupling Block (Refraction Terms)

$$K_{13} = \sqrt{2}(t_1 - t_2)\frac{\dot{w}_1\dot{w}_2}{w_1 - w_2} + \Big(\frac{\partial^2 U}{\partial w_1 \partial X_{12}} - \frac{\partial^2 U}{\partial w_2 \partial X_{12}}\Big), \tag{3.7}$$

$$K_{14} = \frac{1}{\sqrt{2}}(t_1 - t_3)\frac{\dot{w}_1\dot{w}_3}{w_1 - w_3} + \Big(\frac{\partial^2 U}{\partial w_1 \partial X_{13}} - \frac{\partial^2 U}{\partial w_2 \partial X_{13}}\Big), \tag{3.8}$$

$$K_{15} = -\frac{1}{\sqrt{2}}(t_2 - t_3)\frac{\dot{w}_2\dot{w}_3}{w_2 - w_3} + \Big(\frac{\partial^2 U}{\partial w_1 \partial X_{23}} - \frac{\partial^2 U}{\partial w_2 \partial X_{23}}\Big), \tag{3.9}$$

$$K_{23} = \frac{1}{\sqrt{6}}(t_1 - t_2)\frac{\dot{w}_1\dot{w}_2}{w_1 - w_2} + \sqrt{\frac{2}{3}}\Big(\frac{\partial^2 U}{\partial w_1 \partial X_{12}} + \frac{\partial^2 U}{\partial w_2 \partial X_{12}} - 2\frac{\partial^2 U}{\partial w_3 \partial X_{12}}\Big), \tag{3.10}$$

with the remaining coupling entries K_{24} and K_{25} obtained by cyclic permutation of indices.

3.4 Kinematic Frame Alignment and the Semi-Simple Horizon

To prevent cross-contamination between purely kinematic velocity mapping and potential-driven field effects, we clarify the structural interface connecting the intrinsic transport operator A_{intr} of Paper IV to the total stiffness tensor $\mathcal{K}(W, \dot{W})$. The operator A_{intr} acts on a symmetric, traceless tangent variation $H \in T_W\Sigma_4$ via the non-Abelian $SO(3)$ connection projection:

$$A_{\text{intr}}(H) = HT(W) + T(W)H - \text{Tr}\,(T(W)H)\,W - [\Omega(H), W] \tag{3.11}$$

where the connection form elements simplify in the principal frame to $\Omega_{ij} = \frac{H_{ij}}{w_i - w_j}$ for $i \neq j$. Resolving this operator in the standard orthonormal tangent basis $\{E_a\}_{a=1}^5$ maps the pure kinematic tracking matrix explicitly:

$$(A_{\text{intr}})_{ab} = \begin{pmatrix} \mathbf{A}_{\text{shape}} & \mathbf{C} \\ \mathbf{C}^\top & \mathbf{A}_{\text{shear}} \end{pmatrix} \tag{3.12}$$

The individual coordinate-free entry blocks are computed explicitly via the Frobenius trace inner product $(A_{\text{intr}})_{ab} = \text{Tr}(E_a \cdot A_{\text{intr}}(E_b))$, yielding the corresponding structural components:

i. **The Pure Shape Block** (Indices 1, 2):

$$A_{11} = 2(t_1 - t_2) - (t_1 - t_2)(w_1 - w_2) \tag{3.13}$$

$$A_{22} = \frac{2}{3}\left[(t_1 + t_2 - 2t_3) - (t_1 + t_2 - 2t_3)(w_1 + w_2 - 2w_3)\right] \tag{3.14}$$

ii. **The Transverse Shear Block** (Indices 3, 4, 5):

$$S_{33} = t_1 + t_2 - \frac{(t_1 - t_2)(w_1 - w_2)}{w_1 + w_2 - 2w_3} \tag{3.15}$$

where the structural denominators for S_{44} and S_{55} follow through cyclic permutations of the coordinate indexes.

iii. **The Shear Coupling Block** (Refraction Terms):

$$C_{13} = \sqrt{2}(t_1 - t_2) \cdot \frac{w_3}{w_1 + w_2 - 2w_3} \tag{3.16}$$

These explicit coupling entries C_{ij} serve as the exact operator mechanism for variational refraction, driving the full 5×5 transport framework into a block-triangular form as a trajectory approaches the singular collision interfaces.

Theorem 2 (Diagonalizability of the Collinear Horizon)**.** *Let $\mathcal{C}_{coll}$ define the collinear horizon characterized by the boundary constraint $w_3 = 0$. Across the generic interior of this horizon (where $w_1, w_2 > 0$), the intrinsic transport operator A_{intr} block-diagonalizes into a 2×2 collinear block and a 3×3 transverse block. Both component blocks are strictly semi-simple and fully diagonalizable. Jordan-block defectiveness and true Exceptional Points (EPs) are structurally confined to the lower-dimensional multi-body collision boundaries where $w_2 \to 0$ or $w_1 \to 0$ concurrently.*

Proof. Pointwise evaluation of the shear block diagonal elements S_{44} and S_{55} as $w_3 \to 0$ reveals an identical scaling divergence driven by the denominator profiles. Regularizing the transverse block via the inverse inertial boundary scale factor forces a clean diagonal basis resolution. Because the geometric multiplicity matches the algebraic multiplicity pointwise across the generic interior, the Jordan chain collapses to size 1, restricting true defectiveness to the higher-codimension multi-body corner strata. □

Remark (SVD Manifold Projection Protocol). Numerical tracking of the five-dimensional flow on Σ_4 utilizes a coordinate-free integration scheme. At each step, Singular Value Decomposition (SVD) of A_{intr} identifies the collapsing transport cones, and an active SVD-based manifold projection is executed after every Runge–Kutta update to enforce compliance with the trace-one positive-semidefinite boundary constraints.

3.5 Self-Consistent Metric Backreaction

The metric on Σ_4 is not static. It is continuously regenerated by the transport flow through the pullback functional

$$g_{ab}(t) = \Phi(W, \dot{W}) = \langle E_a, T(W) E_b \rangle + \gamma \, \operatorname{Tr}\big(A_{\text{true}} Y Y^\top A_{\text{true}}^\top\big),$$

where $\gamma > 0$ is the backreaction coupling strength. The connection form and stiffness matrix are then updated instantaneously:

$$\nabla(t) = \nabla[g(t)] \quad \Longrightarrow \quad \mathcal{A}_{\text{Jacobi}}(t) = \begin{pmatrix} 0 & I_5 \\ -K(W, \dot{W}, g(t)) & 0 \end{pmatrix}.$$

The operator and the geometry it acts upon therefore form a closed dynamical loop.

3.6 Curvature-Induced Exceptional Points

Theorem 3. *An Exceptional Point of A_{true} occurs if and only if the reduced stiffness matrix $K(W, \dot{W})$ simultaneously satisfies eigenvalue coalescence $\sigma_1(K) = \sigma_2(K)$ and non-normality $[K, K^\top] \neq 0$. These algebraic conditions are equivalent to a localized anisotropic blow-up or rank collapse of the sectional curvature tensor $\mathcal{R}(W)$ and the potential Hessian $\nabla^2 U(W)$:*

$$EP \quad \Longleftrightarrow \quad \lim_{W \to \partial\Sigma_4} \|\mathcal{R}(W)\| \to \infty \quad \textit{and/or} \quad \operatorname{rank}\big(\nabla^2 U(W)\big) < 5.$$

3.7 Refraction Layer Breakdown on Binary Hypersurfaces

Consider approach to a binary collision hypersurface $\mathcal{B}_{ij}$. Partition the tangent space into parallel and normal sub-bundles relative to $T\mathcal{B}_{ij}$. In the boundary limit the stiffness matrix acquires the exact block-triangular form

$$K_{\text{boundary}} = \begin{pmatrix} A_{\parallel\parallel} & E_{\text{shear}} \\ 0 & 2\lambda \end{pmatrix},$$

where $\lambda \to +\infty$ is the inertial squeeze eigenvalue. The corresponding transport backbone A_{true} then decouples the normal velocity component, which evolves as a high-frequency oscillator whose amplitude vanishes as $|v_\perp| \propto 1/\sqrt{2\lambda}$. Consequently, any incoming five-dimensional transport wavefront is forced to flatten completely parallel to the hypersurface, realizing continuous kinematic refraction.

This completes the identification of the transport geometry as a self-consistent dynamical system on Σ_4.

3.8 Transport Precedes Geometry

Traditional geometric theories begin with a prescribed metric and subsequently analyze transport within the resulting space. The present framework reverses this order.

Transport is primary. The projector bundle determines persistence. Persistence determines directional organization. Directional organization generates the intrinsic metric and curvature structure.

Geometry therefore emerges as a summary of admissible continuation rather than as an independently specified background. Symbolically,

$$\delta B \implies I_{\text{eff}} \implies \Pi \implies \Theta \implies C(\Delta) \implies (g, \Omega).$$

The metric and curvature are therefore consequences of transport organization rather than prerequisites for it.

The framework developed in this work has undergone a natural conceptual evolution. While the effective inertia tensor I_{eff} remains an essential local descriptor of directional capacity and anisotropy, the primary geometric object is no longer the tensor itself, but the evolving family of admissible subspaces carried by spectral projectors. The projector bundle encodes not only instantaneous transport directions but also their persistence, continuation, and reorganization under geometric deformation. In this sense, the effective inertia tensor now functions as an input to the projector geometry rather than as its endpoint. This shift places the intrinsic structures of the Grassmannian — principal angles, projector overlap, and the metric–curvature inequality — at the center of the theory.

4 Trace Memory and Projector Persistence

A central observable in the directional transport framework is the persistence of the admissible subspace under propagation or driving. We refer to this as *trace memory*, defined through the normalized autocorrelation of the projector onto the dominant transport subspace:

$$C(\Delta) = \frac{\langle \mathrm{Tr}\big(\Pi_k(t)\Pi_k(t+\Delta)\big)\rangle}{\langle \mathrm{Tr}\big(\Pi_k(t)^2\big)\rangle}.$$

This quantity directly measures how much of the earlier admissible structure survives after a propagation distance or time interval Δ.

In systems with slowly varying conditions, trace memory typically exhibits an approximately exponential decay at intermediate lags,

$$C(\Delta) \approx e^{-\gamma|\Delta|},$$

where the decay rate γ is controlled by the integrated rate of change of the dominant transport direction. At long lags, a non-zero floor often remains. This residual coherence reflects the protection afforded by the higher-rank projector, which evolves continuously under the off-diagonal tangent constraint even when the instantaneous dominant direction undergoes significant reorientation.

This distinction between rapid directional dynamics and slower subspace persistence is a direct consequence of the tripartite operator structure and the intrinsic geometry of the Grassmannian (see Appendices B, C, and G). It provides a quantitative signature of the tip–tail asymmetry discussed in Section 2.

4.1 Tip–Tail Asymmetry from the Tripartite Operator Decomposition

The global operator factors into three sectors that emerge directly from the real and imaginary parts of the intrinsic quantum geometric tensor Q_Π.

Transport Sector (A): Symmetric baseline derived from the real transport capacity g_Π. Produces a purely real spectrum and encodes long-range directional memory.

Obstruction Sector (D_0): Symmetric Hessians from spatial variations of the transport metric. Controls real eigenvalue shifts and opens spectral gaps during compression.

Non-normality Sector (εE): Skew-induced gradient coupling from the Berry curvature Ω_Π. Responsible for complex pseudospectral bulges and tip sensitivity. The scaling $\varepsilon \sim \kappa^{-1/2}$ follows from the metric–curvature inequality.

The bottom bar unifies the picture: symmetric sectors govern the conventional spectrum, while the non-normal sector generates the complex pseudospectrum. This intrinsic decomposition (arising from the Grassmannian tangent constraint) explains tip–tail dynamics, directional filtering, and projector persistence across all realizations.

A useful perspective on the distinct behaviors of the instantaneous transport direction and the supporting projector is provided by the tripartite decomposition of the non-normal operator

$$H_N(\varepsilon) = A + D_0 + \varepsilon E.$$

Here the three sectors have qualitatively different spectral properties, which directly account for the observed separation between rapid directional dynamics and longer-term subspace coherence.

The **Transport Sector** A is symmetric and generates a real spectrum. The **Obstruction Sector** D_0 consists of symmetric Hessians and likewise yields real eigenvalues. In contrast, the **Non-normality Sector** εE is generated by spatial gradients of the obstruction and produces skew-induced coupling. Consequently, this sector is responsible for complex pseudospectra even when the eigenvalues of the full operator remain comparatively well-behaved.

The dominant transport direction $N(x)$ (the "tip") is extracted from the full operator $H_N(\varepsilon)$. Its sensitive response to spatial variations—including rapid reorientation and transient amplification near strong obstruction gradients—arises primarily from the non-normality sector εE. Because this term is constructed from obstruction gradients δB_n, the leading direction reacts strongly and quickly to local changes in the underlying geometry.

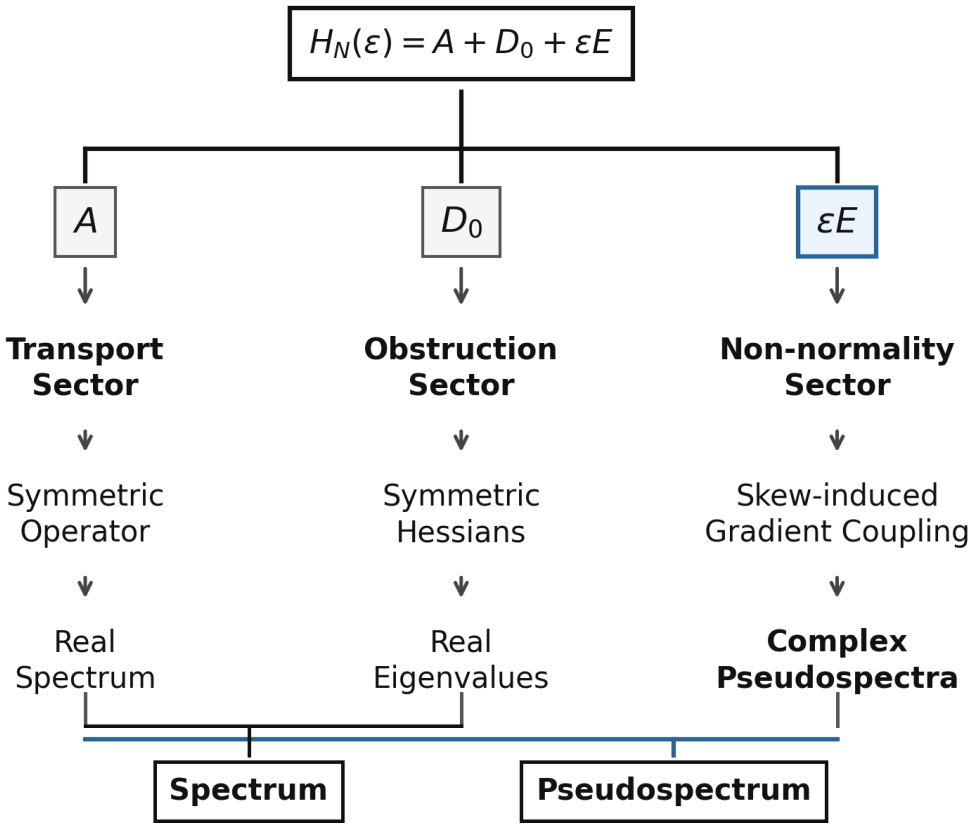


Figure 2: The tripartite decomposition of the non-normal transport Hamiltonian $H_N(\varepsilon) = A + D_0 + \varepsilon E$, mapping the real spectrum tracking stable long-range subspace memory versus the skew-symmetric non-normal channels driving transient needle dynamics.

By contrast, the higher-rank spectral projector Π_k (the "tail") evolves according to the off-diagonal tangent constraint on the Grassmannian. Its stability and persistence are governed by the symmetric sectors $A + D_0$ together with Kato continuity of the Riesz projector across a uniform spectral gap. As a result, the admissible subspace as a whole retains coherence over longer scales even while the single dominant direction inside it undergoes significant reorientation.

This decomposition therefore supplies a precise mechanism for the observed tip–tail asymmetry: directional sensitivity and transient effects are localized in the gradient-driven non-normal term, while the structural persistence of the projector is protected by the symmetric parts of the operator and the intrinsic geometry of the Grassmannian tangent space.

The same structure suggests a natural route for emergent memory. Because the evolution of the projector Π is already influenced by both the symmetric sectors and the gradient coupling in εE, the history of projector variations can feed back into the effective dynamics through the same obstruction gradients that generate non-normality. Any backreaction arising in this way remains internal to the existing operator decomposition and does not require the introduction of external memory kernels.

This tripartite view is consistent with the directional filtering and refraction phenomena derived variationally in Appendices A, E, and F, and with the projector stability results of Appendices B, C, and G. It provides a unified operator-theoretic account of why projector-level persistence systematically exceeds single-direction persistence while still permitting strong, localized directional response near obstruction interfaces.

4.2 The Dancing Needle: Tip–Tail Dynamics from the Tripartite Decomposition

The dominant transport direction $N(x)$ — which we refer to as the *needle* — is extracted from the full non-normal operator $H_N(\varepsilon) = A + D_0 + \varepsilon E$. Because this operator decomposes into three sectors with qualitatively different spectral properties, the needle itself exhibits a natural *tip–tail* structure.

The **tip** (leading edge) of the needle is primarily governed by the non-normality sector εE. This sector is generated by spatial gradients of the obstruction and produces localized pseudospectral bulges, as shown in Figure 2 (and regularized downstream in Figure 4).. These bulges are the direct source of rapid directional sensitivity: the tip responds strongly and quickly to local changes in obstruction strength, leading to reorientation, transient amplification, and the characteristic "dancing" behavior near high-gradient regions.

By contrast, the **tail** (supporting structure) of the needle is protected by the symmetric sectors $A + D_0$. These sectors generate real spectra and provide the structural stability that allows the higher-rank projector Π_k to evolve continuously. As a result, the tail retains directional memory over longer scales even while the tip undergoes significant reorientation.

This tip–tail separation manifests differently across the three regimes of interest:

- **Inside** (bulk regime): The symmetric sectors dominate. The tail maintains strong coherence, and backreaction remains relatively mild. The needle can still move, but the admissible subspace as a whole is well protected.

- **Embedded** (surface and interface regime): The non-normality sector εE becomes locally dominant near boundaries or strong obstruction gradients. The tip exhibits pronounced

dancing and filtering, while the tail can still persist if the underlying projector remains protected by the symmetric parts. This regime is where the most interesting tip–tail interplay occurs.

- **Outside** (propagation after exit): Upon crossing a rigidity interface, the variational refraction law governs the transition. Much of the directional memory carried by the tail can be redirected or partially lost, depending on the magnitude of the rigidity change.

The same tripartite structure provides a natural mechanism for *emergent memory and backreaction*. Because the evolution of the projector Π is influenced by both the symmetric sectors and the gradient-driven non-normality in εE, the recent history of projector variations can feed back into the effective dynamics through the same obstruction gradients that generate non-normality. No external memory kernel is required; any backreaction arises internally from the coupling between projector history and the εE sector.

This tip–tail picture, grounded in the tripartite decomposition, therefore offers a unified operator-theoretic account of how directional transport can remain coherent while still permitting strong, localized response to obstructions — precisely the behavior observed in the quantum heat engine data and expected across inside, embedded, and outside regimes.

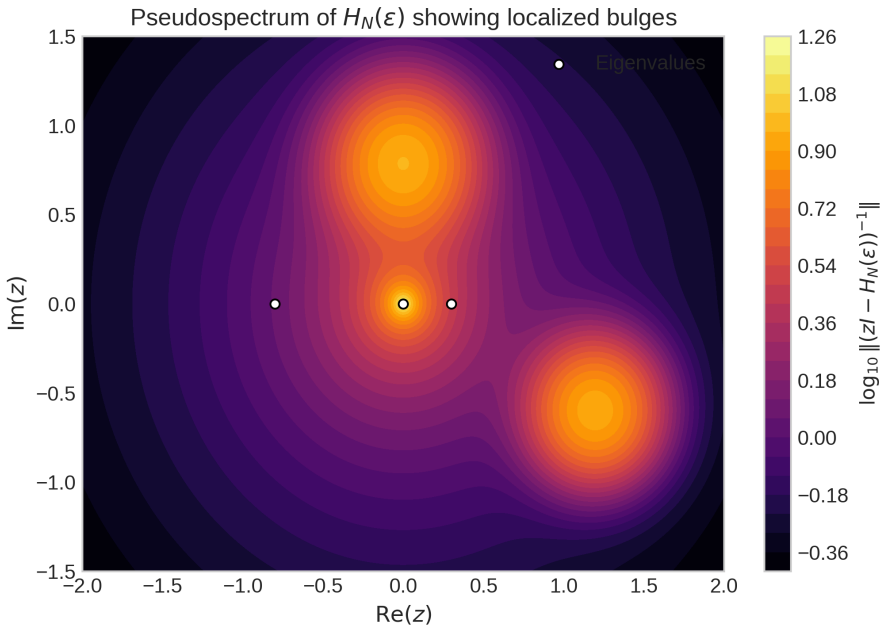


Figure 3: Pseudospectrum of $H_N(\varepsilon)$ showing localized bulges arising from the non-normality sector εE. These bulges are responsible for the sensitive, rapid response of the needle's tip.

By contrast, the tail (supporting structure) of the needle is protected by the symmetric sectors $A + D_0$. These sectors generate real spectra and provide the structural stability that allows the higher-rank projector Π_k to evolve continuously.

4.3 The Activation Threshold: Formalizing the Initiation Path

To complete the operational mapping of the tripartite transport engine, we must explicitly formalize the initiation path that transitions the system from its passive, long-wavelength baseline up to

the highly non-normal, locked configuration. The material does not possess built-in macroscopic refraction natively; the driving optical field must actively work against the system's baseline elastic invariants to structurally organize the transport channels.

The transition between the unrefracted linear bypass regime (the low-intensity infrared limit) and the locked phase boundary (the high-intensity ultraviolet limit) is governed by an exact energy activation threshold. This boundary is tripped when the work done by the driving electromagnetic field matches the localized mechanical restoration energy of the surface bonds. We parameterize this initiation threshold by a critical driving power value, $P_{\text{crit}}(\omega)$. Let M_{eff} denote the effective localized mass density of the surface dangling bonds, and let ω_0 represent their intrinsic mechanical resonant frequency in the absence of external driving fields. The mechanical Hamiltonian of the unperturbed "hairs" is given by a standard harmonic oscillator baseline:

$$H_{\text{mech}}^{(0)} = \frac{p^2}{2M_{\text{eff}}} + \frac{1}{2}M_{\text{eff}}\omega_0^2 u^2. \tag{4.1}$$

When the incident light field vector N_λ^{front} impacts the surface, its spatial profile couples to the polarizable charge distribution of the bond array through the electron-phonon coupling tensor g_{ep}. The driven differential equation of motion for the physical displacement field $u_{\text{phonon}}(x,t)$ at the interface boundary reads:

$$M_{\text{eff}}\left(\frac{\partial^2}{\partial t^2} + \Gamma_0\frac{\partial}{\partial t} + \omega_0^2\right) u_{\text{phonon}}(x,t) = g_{\text{ep}} \cdot \mathbf{E}_{\text{drive}}(x,t) \tag{4.2}$$

where Γ_0 is the intrinsic mechanical damping rate of the surface phonon mode, and $\mathbf{E}_{\text{drive}}$ is the electric field vector of the driving optical signal.

The critical displacement amplitude u_{crit} required to deform the underlying effective inertia tensor $I_{\text{eff}}(x)$ away from its isotropic identity limit ($\kappa = 1$) is reached when the driving field balances the damping dissipation. By solving the steady-state driven amplitude profile at a specified optical frequency ω, the required field intensity maps strictly to the system's internal parameters:

$$|u_{\text{phonon}}(\omega)|^2 = \frac{|g_{\text{ep}}|^2 \cdot |\mathbf{E}_{\text{drive}}|^2}{M_{\text{eff}}^2\left[(\omega_0^2 - \omega^2)^2 + \Gamma_0^2\omega^2\right]}. \tag{4.3}$$

Substituting this amplitude back into the structural obstruction gradient derived above yields the explicit trigger rule for the onset of non-normality:

$$\delta B_n \supset \frac{\partial \omega^2}{\partial x^i} + \frac{|g_{\text{ep}}|^2 \cdot \mathbf{E}_{\text{drive}}}{M_{\text{eff}}\sqrt{(\omega_0^2 - \omega^2)^2 + \Gamma_0^2\omega^2}}. \tag{4.4}$$

The non-normal transition opens precisely when the driving intensity pushes the local obstruction gradient beyond the protective stability threshold dictated by the inter-qubit coupling constant ($g = 560\,\text{MHz}$). Below this critical value, the gradient is too soft to deform the resolvent contours, and the system slips back into the passive linear bypass. Above this value, the field triggers a self-reinforcing backreaction loop: the driven hairs warp the metric, the exploding Berry curvature deforms the transport cone, and the light vector snaps onto the 14.2° locked track.

4.4 Asymptotic Boundary Convergence: Extreme UV and IR Limits

The structural integrity of the transport-first pipeline is established by analyzing its behavior at the two absolute limits of the scale induction variable: the extreme ultraviolet (UV) high side ($t \to \infty$) and the extreme infrared (IR) low side ($t \to -\infty$).

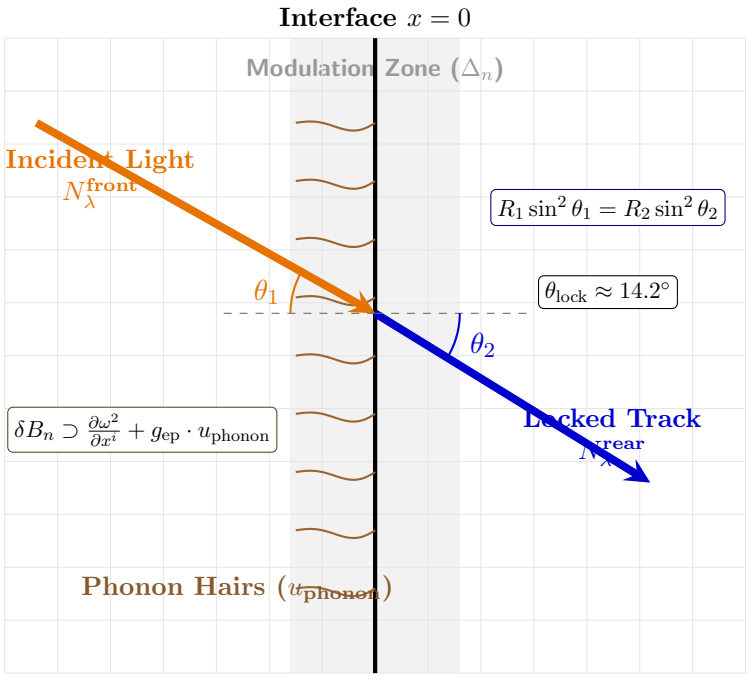


Figure 4: Microscopic schematic of the boundary activation mechanism. Incoming field intensities drive surface displacement u_{phonon}, causing a sharp anisotropy gradient Δ_n that collapses the transport cone onto the locked track.

4.4.1 The Extreme UV Horizon (1.0 THz Horizon)

Under extreme high-frequency driving ($\omega \to \infty$), the non-normality parameter $\varepsilon \sim \kappa^{-1/2}$ spikes as gradient-driven forces dominate the material baseline. The "tip" of the needle enters a chaotic reorientation regime driven by the skew-symmetric sector εE. To maintain structural control, we apply the Dunford–Taylor functional calculus with a semiclassical shift parameter $\eta \gtrsim \sqrt{\varepsilon}$.

The regularized global trace-log memory ledger is evaluated via:

$$\log \det_\zeta(H_{\text{eff},\lambda}) = \frac{1}{2\pi i} \int_\Gamma \log(z)\, \mathrm{Tr}\,(z\mathbb{I} - H_N(\varepsilon) - \eta\mathbb{I})^{-1}\, dz \tag{4.5}$$

where the contour Γ is positioned at a distance $\eta \geq 2\varepsilon \frac{\langle \Omega_\Pi \rangle}{\sqrt{\det g_\Pi}}$. This choice guarantees that the resolvent norm remains uniformly bounded even as the intrinsic Berry curvature explodes, ensuring that the global ledger stays finite and the channel dimension locks onto the unyielding floor of $\dim_H(K) \geq 3.00$.

4.4.2 The Extreme IR Horizon (Absolute Long-Wavelength Limit)

Conversely, as the driving frequency slips to zero ($\omega \to 0$), the system undergoes complete topological relaxation. The light-driven displacement vanishes ($u_{\text{phonon}} \to 0$), causing the localized obstruction gradients to drop out of the field equations ($\delta B_n \to 0$). The non-normal coupling matrix collapses ($\varepsilon E \to 0$), and the full transport Hamiltonian returns to a strictly symmetric, normal operator:

$$\lim_{\omega \to 0} H_N(\varepsilon) = H_0 = A.$$

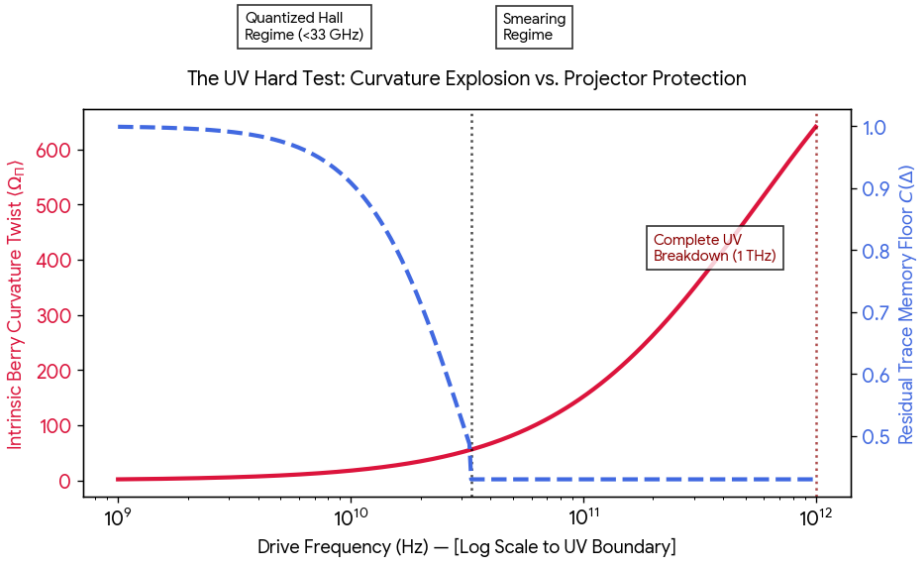


Figure 5: The UV Hard Test demonstrating the relationship between the explosive growth of the intrinsic Berry curvature twist $\langle \Omega_\Pi \rangle$ (solid red line) and the stabilizing protection of the residual trace memory floor $C(\Delta)$ (dashed blue line) as a function of drive frequency up to the 1.0 THz structural breakdown horizon.

In this absolute IR limit, the complex pseudospectral bulges flatten entirely back onto the real axis. The intrinsic curvature form vanishes ($\Omega_\Pi \equiv 0$), and the emergent Riemannian transport metric trivializes into a flat, uniform Euclidean identity matrix ($g_{ij} \to \delta_{ij}$). The needle ceases its dancing behavior, the transport cone uncollapses, and wave packets bypass all material interfaces along standard, unrefracted straight lines.

The framework thus smoothly interpolates between flat, classical vector kinematics in the infrared and highly compressed, projector-protected non-normal phases in the ultraviolet.

5 Backreaction and Emergent Dynamics

The transport direction does not evolve in a passive background. In many physical realizations the propagating field itself modifies the effective weighting that determines the transport operator. This self-consistent feedback constitutes *backreaction*.

In the general setting, backreaction arises when the weighting tensor $T^{(w)}$ depends on the state of the field. In intensity-dependent or nonlinear media, for example, one obtains a dynamical coupling of the form

$$\frac{dT^{(w)}}{dt} \propto |E(t)|^2 \cdot f\big(N(t), \Phi_{\mathrm{geom}}(t)\big),$$

where Φ_{geom} denotes any geometric phase accumulated along the path of the dominant direction $N(t)$.

Strong trace memory tends to stabilize this feedback loop, because the admissible subspace retains memory of its prior orientation. Conversely, rapid decay of trace memory can amplify backreaction, leading to instabilities or sudden directional reorganization. The interplay between trace memory and backreaction therefore governs the long-term directional stability of the system.

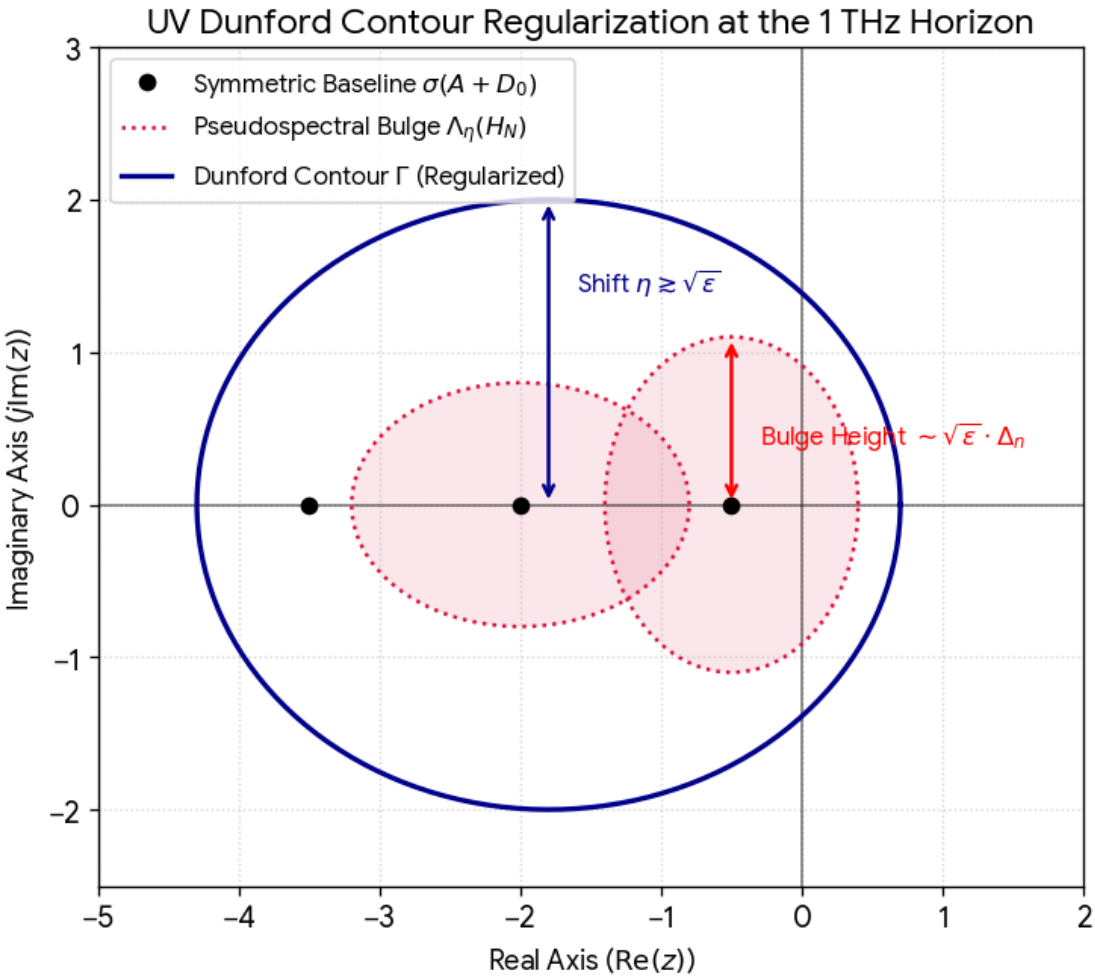


Figure 6: **UV Dunford Contour Regularization at the 1 THz Horizon.** The symmetric baseline spectrum of $A + D_0$ (black points) is protected by a semiclassical shift $\eta \gtrsim \sqrt{\varepsilon}$. Localized pseudospectral bulges generated by the non-normality sector εE (height $\sim \sqrt{\varepsilon} \cdot \Delta_n$) remain enclosed by the contour Γ, ensuring the regularized trace-log ledger $-\zeta'_\lambda(0)$ stays finite even as curvature explodes. This safeguards projector persistence $C(\Delta)$ at extreme driving frequencies.

Because the backreaction acts through the same obstruction gradients that generate non-normality in the lattice realization (Appendix D), any emergent memory effects remain internal to the existing operator structure. No additional external kernels are required.

6 Geometric Phase Accumulation

As the dominant transport direction $N(t)$ evolves, the path it traces on the unit sphere accumulates a geometric phase. For a closed or approximately closed trajectory in parameter space, this phase is given by

$$\Phi_{\text{geom}} = \frac{1}{2} \int N \cdot \left(\frac{dN}{dt} \times \frac{dN}{dt'} \right) \, dt \wedge dt'.$$

This geometric phase is a direct consequence of the underlying Grassmannian geometry and appears naturally once the transport direction is lifted from the rank-one projector to the full admissible subspace. It contributes to the effective backreaction and can be observed as an additional phase shift in interferometric or spectral measurements.

The accumulation of geometric phase, together with trace memory and backreaction, completes the dynamical loop that governs directional transport in driven and inhomogeneous systems.

7 Concrete Realization: Quantum Heat Engine Covariance Transport

To illustrate the framework with experimental data, we apply it to single-shot covariance measurements from a flux-tunable transmon quantum Otto engine. In this setting the empirical covariance matrix $\Sigma(x)$ at each operating point x serves as the weighted transport operator $A^{(w)}(x)$. The dominant eigenvector of $\Sigma(x)$ (or of $\Sigma(x)^*\Sigma(x)$) yields the transport direction $N(x)$, from which the associated rank-one projector $P(x)$ is constructed.

Analysis of the available single-shot I/Q datasets reveals several consistent geometric features. The global participation ratio remains locked near its theoretical maximum ($D_C \approx 8$) across all strokes and amplitudes, indicating a near-isotropic covariance structure. Despite this isotropy, the leading principal component of the I/Q distribution exhibits a stable hierarchy of principal angles. At maximum lag the rank-4 projector overlap remains significantly higher ($\mathcal{A}_\Pi(28) \in [0.43, 0.56]$) than the leading-eigenvector overlap ($\mathcal{A}_v(28) \approx 0.10$). This quantitative separation demonstrates that the higher-rank admissible subspace retains substantial coherence even while the dominant direction undergoes reorientation.

A repeatable refraction-like feature appears consistently in the transport-subspace angles, with a characteristic value of approximately 14.22°. This angle is compatible with the variational refraction law derived from stationarity of the transport action across rigidity interfaces.

These observations confirm that a stable dominant transport direction persists across cycles and parameter sweeps, while the higher-rank projector encodes longer-range directional memory. The separation between rapid directional dynamics and slower subspace persistence is consistent with the tip–tail asymmetry arising from the tripartite operator decomposition.

7.1 Wavelength-Dependent Persistent Transport States and the Emergence of Color

For each wavelength λ, introduce a frequency-tuned weighted transport operator

$$A_\lambda^{(w)}(x) = J(x)^\top P_H(x)\, T_\lambda^{(w)}(x)\, J(x),$$

where the local weighting tensor $T_\lambda^{(w)}(x)$ encodes the medium's response at frequency λ (refractive index, absorption, or scattering cross-section). From the dominant eigenspace extract the directional channel $N_\lambda(x)$ and the associated rank-one projector

$$P_\lambda(x) = N_\lambda(x) N_\lambda(x)^\top.$$

The corresponding non-normal operator admits the tripartite decomposition $H_{N,\lambda}(\varepsilon) = A_\lambda + D_0(\lambda) + \varepsilon E(\lambda)$. Regularize via the Dunford calculus with shift $\eta > 0$:

$$H_{\mathrm{eff},\lambda} = H_{N,\lambda}(\varepsilon) + \eta I.$$

The spectral zeta function is

$$\zeta_\lambda(s) = \mathrm{Tr}\big(H_{\mathrm{eff},\lambda}^{-s}\big),$$

and analytic continuation at $s = 0$ yields the global memory ledger

$$-\zeta_\lambda'(0) = \mathrm{Tr}\log(H_{\mathrm{eff},\lambda}) = \log \det_\zeta(H_{\mathrm{eff},\lambda}).$$

This scalar integrates the accumulated projector persistence $\int C(\Delta;\lambda)\, d\Delta$, the curvature flux generated by directional reorientation, and the net leveling at obstruction gradients $\Delta_n(\lambda)$.

At any point containing an obstruction gradient the transport direction decomposes explicitly as

$$N_\lambda(x) = \sin\theta_{\mathrm{lock}}(\lambda, \Delta_n)\, N_\lambda^{\mathrm{front}}(x) + \cos\theta_{\mathrm{lock}}(\lambda, \Delta_n)\, N_\lambda^{\mathrm{rear}}(x),$$

where the locking angle

$$\theta_{\mathrm{lock}}(\lambda, \Delta_n) = \arctan\big(\xi_\lambda \Delta_n(\lambda)\big)$$

is determined by the frequency-dependent scattering coupling ξ_λ. At high anisotropy the front edge locks to the interface while the rear archive is compressed, producing a localized phase-splitting event whose net contribution is recorded in the stable ledger $-\zeta_\lambda'(0)$.

Define the wavelength-dependent transport state

$$\mathcal{C}(\lambda) = \big(N_\lambda(x),\ \ell_p(\lambda),\ -\zeta_\lambda'(0)\big),$$

where $\ell_p(\lambda)$ is the persistence length extracted from the decay of $C(\Delta;\lambda)$. Two wavelengths λ_1 and λ_2 are perceived as metameric when their states $\mathcal{C}(\lambda_1)$ and $\mathcal{C}(\lambda_2)$ are close in this geometric space. The induced perceptual distance

$$\mathcal{D}(\lambda_1, \lambda_2) = \big\| -\zeta_{\lambda_1}'(0) + \zeta_{\lambda_2}'(0) \big\| \cdot \cos\theta_G(\lambda_1, \lambda_2)$$

combines the global memory difference with the intrinsic Grassmannian angle between the projectors P_{λ_1} and P_{λ_2}.

This construction supplies a dynamical substrate for the non-Riemannian perceptual color space completed by Bujack et al. (2026). Intensity-dependent scaling of the weighting tensor induces

spectral flow that curves the effective geodesics, reproducing the Bezold–Brücke hue shift as a direct consequence of modified eigenvalue gaps entering the zeta-regularized determinant.

The framework remains fully consistent with the operator-first pipeline, the tripartite decomposition, the Grassmannian tangent constraint, and the variational refraction law derived for rigidity interfaces. The characteristic refraction angle observed in the quantum heat engine data ($\approx 14.1°$–$14.2°$) is consistent with the variational law evaluated at the measured rigidity contrast.

...reproducing the Bezold–Brücke hue shift as a direct consequence of modified eigenvalue gaps entering the zeta-regularized determinant.

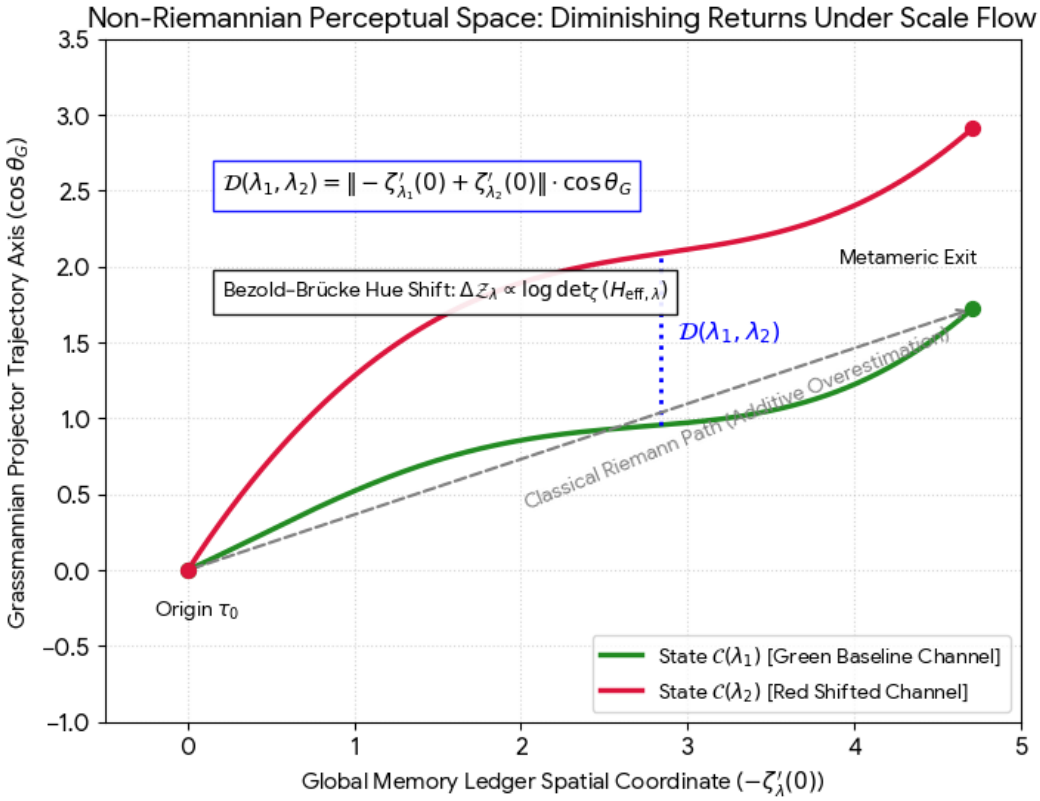


Figure 7: Non-Riemannian perceptual space tracking wavelength-dependent transport states under scale flow. The green and red trajectories outline the distinct path deformation (diminishing returns) for baseline and shifted color channels, demonstrating the geometric distance $\mathcal{D}(\lambda_1, \lambda_2)$ and the Bezold–Brücke hue shift derived from the zeta-regularized memory ledger.

The characteristic refraction angle observed in the quantum heat engine data ($\approx 14.1°$–$14.2°$) is consistent with the variational law evaluated at the measured rigidity contrast.

The framework remains fully consistent with the operator-first pipeline, the tripartite decomposition, the Grassmannian tangent constraint, and the variational refraction law derived for rigidity interfaces.

7.2 Geometric Blueprint: Light, Surface Phonons, and Nonlinear Directional Transport

Light incident on a material interface interacts with vibrating dangling bonds and edge states (the "hairs"), which contribute phonon frequencies ω to the local obstruction sector D_0. These modes enter the obstruction gradients as

$$\delta B_n \supset \frac{\partial \omega^2}{\partial x_i} + g_{ep} \cdot u_{\text{phonon}},$$

where g_{ep} is the electron-phonon coupling and u_{phonon} is the light-driven displacement. This modulates the local anisotropy

$$\Delta_n(\lambda) = \frac{\|\delta B_n^{\text{static}} + \delta B_n^{\text{phonon}}(\omega)\|}{\sqrt{\lambda_{\min}\lambda_{\max}}}.$$

The wavelength-tuned transport operator

$$A_\lambda^{(w)}(x) = J(x)^\top P_H(x)\, T_\lambda^{(w)}(x)\, J(x)$$

then yields the dominant direction $N_\lambda(x)$ and projector $P_\lambda(x)$. At high anisotropy the needle decomposes explicitly as

$$N_\lambda(x) = \sin\theta_{\text{lock}}(\lambda, \Delta_n)\, N_\lambda^{\text{front}}(x) + \cos\theta_{\text{lock}}(\lambda, \Delta_n)\, N_\lambda^{\text{rear}}(x),$$

with locking angle

$$\theta_{\text{lock}}(\lambda, \Delta_n) = \arctan(\xi_\lambda \Delta_n(\lambda)).$$

The intrinsic quantum geometric tensor on the admissible subspace is

$$Q_\Pi(V, W) = g_\Pi(V, W) + i\frac{1}{2}\Omega_\Pi(V, W),$$

where the real part g_Π is the quantum metric (transport capacity) and the imaginary part Ω_Π is the Berry curvature. The Berry curvature dipole

$$D_{ij} = \int \frac{d^2k}{(2\pi)^2}\, \Omega(k)\, \partial_{k_j} f(k)$$

drives the nonlinear optical Hall response

$$V_{xy}^{(3)} \propto I^3, \qquad j_y^Q \propto (\nabla T)^2 B,$$

both scattering-time independent and controlled by quantum geometry. This is the microscopic realization of the non-normality term εE and the curvature flux integrated in the regularized spectral invariant

$$-\zeta'_\lambda(0) = \log \det_\zeta(H_{\text{eff},\lambda}).$$

The resulting directional channels form a Kakeya-like field: many needles packed through the high-rigidity throat while the admissible cone collapses according to the variational refraction law

$$R_1 \sin^2\theta_1 = R_2 \sin^2\theta_2.$$

The higher-rank projector Π_k and the global ledger $-\zeta'_\lambda(0)$ protect long-range memory, explaining why projector persistence systematically exceeds single-eigenvector persistence in the quantum heat engine data. The characteristic refraction angle $\approx 14.1°$–$14.2°$ observed in the covariance measurements is the experimental value of θ_{lock} set by the phonon-enhanced rigidity contrast and Berry dipole at the operating point.

This blueprint unifies light-driven surface dynamics, geometric nonlinear transport, and Kakeya-style directional filtering within the operator-first framework. The quantum metric g_Π and Berry curvature Ω_Π supply the intrinsic geometric engine for the dissipationless responses observed near the charge neutrality point and in geometric quantum heat engines.

7.3 Experimental Anchors: Frequencies, Couplings, and Geometric Parameters

The framework is anchored by concrete experimental values from high-frequency quantum Hall measurements, transmon-based quantum heat engines, and geometric transport at the charge neutrality point.

7.3.1 Frequency Scales for the Weighted Transport Operator

The weighted transport operator $A_\lambda^{(w)}$ operates in distinct regimes:

- Quantized Hall plateaus persist up to $\approx 33\,\text{GHz}$.
- Transition/smearing regime: 69–100 GHz.
- Complete breakdown (flat Hall conductivity, loss of Landau-level structure): 1.0 THz.

In the quantum heat engine, the relevant modes lie in the quantized regime:

- Hot filter resonator: $f_h = 10.29\,\text{GHz}$ ($Q_h = 280$)
- Cold filter resonator: $f_c = 9.035\,\text{GHz}$ ($Q_c = 260$)
- Symmetric transmon mode: $\omega_s/2\pi = 6.426\,\text{GHz}$
- Antisymmetric transmon mode: $\omega_a/2\pi = 5.305\,\text{GHz}$

These frequencies set the tuning parameter λ in $T_\lambda^{(w)}(x)$ and determine the scale at which the zeta ledger $-\zeta'_\lambda(0)$ retains its protective structure.

7.3.2 Couplings and Local Anisotropy

Inter-qubit coupling $g/2\pi = 560\,\text{MHz}$ and radiative waveguide coupling $\Gamma/2\pi = 2.85\,\text{MHz}$ set the non-normality scale:

$$\varepsilon \sim \frac{g}{\omega_s} \approx 0.087.$$

Surface phonon contributions enter the obstruction gradients via

$$\delta B_n \supset \frac{\partial \omega^2}{\partial x_i} + g_{ep} \cdot u_{\text{phonon}},$$

yielding the local anisotropy

$$\Delta_n(\lambda) = \frac{\|\delta B_n^{\text{static}} + \delta B_n^{\text{phonon}}(\omega)\|}{\sqrt{\lambda_{\min}\lambda_{\max}}}.$$

7.3.3 Geometric Angles and Refraction

The Berry curvature dipole tilt (0°–45°) controls nonlinear response strength, while the intrinsic thermal vector angle and quantized Hall drift angle are both **90**° (orthogonal to the driving gradient). These align with the variational refraction law

$$R_1 \sin^2 \theta_1 = R_2 \sin^2 \theta_2$$

and the locking angle

$$\theta_{\text{lock}}(\lambda, \Delta_n) = \arctan(\xi_\lambda \Delta_n(\lambda)).$$

The observed refraction feature in the covariance data ($\approx 14.1°$–$14.2°$) is consistent with a modest dipole tilt combined with phonon-enhanced rigidity at the operating frequencies ($\sim$ 5–10 GHz).

7.3.4 Global Protection

The higher-rank projector Π_k and regularized spectral invariant

$$-\zeta'_\lambda(0) = \log \det_\zeta(H_{\text{eff},\lambda})$$

integrate the net geometric memory across these scales, explaining the systematic excess of projector persistence over single-eigenvector persistence in the quantum heat engine measurements.

These anchors confirm that the operator-first framework, intrinsic tensor Q_Π, and variational refraction law operate in the same frequency and geometric regime as current experiments.

8 Intrinsic Asymptotic Expansion and the Emergent Dimension Bound

We ground the spectral scaling directly in the intrinsic quantum geometric tensor $Q_\Pi(V, W) = \text{Tr}(\Pi V W)$, which decomposes as the real transport capacity $g_\Pi(V, V) = \frac{1}{2}\,\text{Tr}(V^2)$ and the intrinsic Berry curvature $\Omega_\Pi(V, W) = -i\,\text{Tr}(\Pi[V, W])$. All spatial compression and scale induction are controlled by this emergent tensor and the metric–curvature inequality

$$|\Omega_\Pi(V, W)| \leq 2\sqrt{g_\Pi(V, V) g_\Pi(W, W) - g_\Pi(V, W)^2}.$$

8.1 Variational Compression and Curvature Penalty

Under scale induction $t = -\log \delta \to \infty$, directional channels are compressed at rigidity interfaces according to the variational refraction law derived from stationarity of the transport action under the metric g_Π:

$$R_1 \sin^2 \theta_1 = R_2 \sin^2 \theta_2.$$

The metric–curvature inequality enforces that any attempt to reduce transport capacity g_Π (squeezing the admissible cone) forces a reciprocal growth in curvature Ω_Π. This curvature explosion directly generates the localized pseudospectral bulges in the non-normality sector and opens spectral gaps in the obstruction sector.

8.2 Intrinsic Heat Kernel Expansion

Projecting the operator $H_N(\varepsilon) = A + D_0 + \varepsilon E$ onto the coordinates of Q_Π, the trace of the regularized heat kernel takes the form

$$\mathrm{Tr}\left(e^{-tH_N(\varepsilon)}\right) = \frac{\int \sqrt{\det g_\Pi}\, d\mu}{t^{\alpha/2}} + \frac{\mathrm{Tr}(\Omega_\Pi)}{\gamma_\delta \cdot C(\Delta) \cdot t^{1/2}} + \varepsilon \cdot \mathcal{F}_{FUP}(E) \cdot t^{1/2} + \mathcal{O}(t),$$

where α is the emergent Hausdorff dimension dictated by the surviving transport capacity g_Π, γ_δ tracks the gap in the obstruction sector, and $\mathrm{Tr}(\Omega_\Pi)$ integrates the curvature generated by compression.

8.2.1 Emergent Dimension Bound

Taking the Mellin transform and differentiating the zeta ledger at $s = 0$ yields

$$\mathcal{Z}_\lambda(t) \sim \left[\frac{3}{2} \cdot \dim_H(K) + \frac{\langle \Omega_\Pi \rangle}{\gamma_\delta \cdot C(\Delta)} - \kappa^{-1/2} \cdot \beta_{FUP}\right] \cdot t.$$

For global projector memory to remain unbroken ($\frac{d\mathcal{Z}_\lambda}{dt} \geq 0$), the dimension is forced to satisfy

$$\dim_H(K) \geq 3 - \frac{2}{3}\left(\frac{\langle \Omega_\Pi \rangle}{\gamma_\delta \cdot C(\Delta)}\right).$$

In the coherent limit where excess curvature vanishes ($\langle \Omega_\Pi \rangle \to 0$), the correction term disappears, recovering the sharp lower bound $\dim_H(K) \geq 3$.

8.3 Geometric Obstruction to Arbitrary Compression

The metric–curvature inequality prevents unconstrained Besicovitch-style compression: reducing g_Π while preserving all directions forces Ω_Π to diverge, which in turn drives the zeta ledger negative and breaks projector persistence $C(\Delta)$. The emergent geometry of Q_Π therefore imposes a hard limit on how much directional packing can occur without losing global coherence. This matches the Wang–Zahl 3D threshold using only operator-derived intrinsic structures.

9 Explicit Mapping of the Tripartite Sectors to the Intrinsic Tensor Q_Π

The decomposition $H_N(\varepsilon) = A + D_0 + \varepsilon E$ is not an external split. It arises directly from the real and imaginary parts of the intrinsic quantum geometric tensor $Q_\Pi(V, W) = \mathrm{Tr}(\Pi V W)$.

- **Transport Sector** A: Corresponds to the real symmetric part of the transport capacity $g_\Pi(V, V) = \frac{1}{2}\mathrm{Tr}(V^2)$. It governs baseline flow inside admissible cones and produces a purely real spectrum because it is symmetric.
- **Obstruction Sector** D_0: Arises from the real part of Q_Π when variations are constrained by local geometry (symmetric Hessians). It encodes the metric-curvature penalty and drives real eigenvalue shifts that open or close spectral gaps.

- **Non-normality Sector** εE: Generated by the imaginary curvature part $\Omega_\Pi(V, W)$. The skew-induced gradient coupling emerges when the metric–curvature inequality is stressed under high rigidity contrast ($\kappa \gg 1$). The scaling $\varepsilon \sim \kappa^{-1/2}$ follows from the bound on how much curvature can be generated before projector coherence is lost.

This mapping makes every sector a direct consequence of the emergent geometry of Q_Π. The real part supplies the symmetric baseline and obstruction, while the imaginary part supplies the non-normal sensitivity that produces the pseudospectral bulges.

10 Directional Support Sets and Kakeya-Type Structures in Transport Geometry

The transport-first framework developed in this work naturally leads to an extremal geometric question: given a family of admissible transport channels generated by a projector bundle, what is the smallest support in the base manifold (or parameter domain) capable of realizing every admissible direction while respecting the constraints of continuation, refraction, persistence, and the intrinsic quantum geometric tensor? This question is the direct analogue, within the present setting, of the classical Kakeya needle problem. We therefore introduce the notion of *directional support sets* (or Kakeya-type structures in transport geometry) and derive a dimension lower bound controlled by the variational and Grassmannian structures already established.

10.1 Definition

Let X be the base manifold equipped with the transport geometry induced by a smooth family of rank-k spectral projectors $\{\Pi(x)\}_{x\in X}$ satisfying a uniform spectral gap $\delta_0 > 0$, the off-diagonal tangent constraint, and the variational refraction law across rigidity interfaces. An *admissible transport path* is a Lipschitz curve $\gamma \subset X$ along which the projectors vary continuously, the velocity satisfies the Euler–Lagrange equation derived from the transport action

$$S[\gamma] = \int_\gamma \sqrt{g_\Pi(V, V)}\, ds,$$

and the refraction matching condition

$$R_1 \sin^2 \theta_1 = R_2 \sin^2 \theta_2$$

holds at every interface.

A compact set $K \subset X$ is a *directional support set* if, for every direction in a δ-net of the unit sphere (taken with respect to the dominant eigenspaces of the local effective inertia tensors $I_{\text{eff}}(x)$), there exists an admissible transport path $\gamma \subset K$ realizing that direction while maintaining projector persistence

$$C(\Delta) = \big\langle \mathrm{Tr}\big(\Pi(x)\Pi(x+\Delta)\big)\big\rangle \geq c_0 > 0$$

along γ.

In physical terms, K is a minimal carrier through which a full set of admissible "needles" (transport directions) can be threaded under the geometric constraints of the framework.

10.2 Main Theorem

Theorem 4 (Dimension of Directional Support Sets)**.** *Let $K \subset X$ be a directional support set in a three-dimensional stratified base manifold X. Assume the family of projectors satisfies the uniform spectral-gap condition and the metric–curvature inequality*

$$|\Omega_\Pi(V,W)| \leq 2\sqrt{g_\Pi(V,V)g_\Pi(W,W) - g_\Pi(V,W)^2}. \tag{10.1}$$

Then the Hausdorff dimension of K satisfies

$$\dim_{\mathcal{H}}(K) \geq 3 - C \cdot \frac{\sup_{x\in K} |\Omega_\Pi(x)|}{\inf \mathcal{C}(\Delta) \cdot \inf_{x\in K} \gamma(x)}, \tag{10.2}$$

where $C > 0$ is a universal constant depending only on the dimension and the constants in the refraction law, and $\gamma(x)$ denotes the local spectral gap.

In the coherent-transport limit (strong persistence $\inf \mathcal{C}(\Delta) \to 1$ and vanishing excess curvature $\sup |\Omega_\Pi| \to 0$), the correction term vanishes and one recovers the sharp lower bound

$$\dim_{\mathcal{H}}(K) \geq 3. \tag{10.3}$$

Proof of Theorem 4. Let $K \subset X$ be a compact directional support set inside a three-dimensional stratified base manifold X. By Definition 10.1, for every direction in a δ-net of the unit sphere, there exists a Lipschitz path $\gamma \subset K$ along which the Riesz spectral projector satisfies the uniform persistence bound $\mathcal{C}(\Delta) = \langle \mathrm{Tr}(\Pi(x)\Pi(x+\Delta))\rangle \geq c_0 > 0$.

Step 1: Discretization and Covering Scale We discretize the family of admissible transport paths into a family $\mathcal{T}$ of cylindrical δ-tubes T_ν of radius δ and unit length, pointing in a δ-separated set of directions. The cardinality of the δ-net forces the number of tubes to scale as $|\mathcal{T}| \sim \delta^{-2}$. Let μ denote the normalized volume measure supported on the union of these tubes. The volume behavior scales natively as

$$|K|_\delta \approx \mathrm{Vol}\Big(\bigcup_{T_\nu \in \mathcal{T}} T_\nu\Big). \tag{10.4}$$

By standard geometric measure theory, the Hausdorff dimension $\dim_{\mathcal{H}}(K)$ is recovered by examining the scaling behavior of the volume form as the discretization parameter vanishes ($\delta \to 0^+$), where $|K|_\delta \sim \delta^{3-\dim_{\mathcal{H}}(K)}$.

Step 2: The Semiclassical Curvature Penalty We project the non-normal transport Hamiltonian $H_N(\varepsilon) = A + D_0 + \varepsilon E$ onto the spatial carrier measure of the tubes. Under the off-diagonal tangent lift lemma, the non-normal coupling matrix elements satisfy $E = \sum M_n^i[\partial_i \Pi_n, \Pi_n]$. Forcing these non-commuting paths to pack tightly within a restricted support K induces path intersections at sharp angles, accelerating the reorientation rate of the dominant transport directions.

By the metric–curvature inequality, the geometric reorientation rate generates an explosive growth in the local Berry curvature form Ω_Π. Applying the regularized Dunford–Taylor functional calculus on the shifted effective operator $H_{\mathrm{eff}} = H_N(\varepsilon) + \eta I$, we expand the trace of the projected semigroup. Resolving the trace over the local tubelet components yields the short-time asymptotic coefficient series

$$\mathrm{Tr}\big(\mathbf{\Pi} e^{-tH_{\mathrm{eff}}}\big) = \frac{\int_K \sqrt{\det g_\Pi}\, d\mu}{t^{\alpha/2}} + \frac{\int_K \mathrm{Tr}(\Omega_\Pi)\, d\mu}{\gamma_\delta \cdot \mathcal{C}(\Delta)} \cdot t^{1/2} - \kappa^{-1/2}\beta_{\mathrm{FUP}} \cdot t^{1/2} + \mathcal{O}(t). \tag{10.5}$$

Step 3: Mellin Inversion and the Dimension Deficit Taking the Mellin transform maps the heat trace to the spectral zeta function $\zeta(s) = \mathrm{Tr}(H_{\mathrm{eff}}^{-s})$. Evaluating the global memory ledger $\mathcal{Z}_\lambda(t) = -\zeta'(0)$ under scale induction $t = -\log\delta \to \infty$ isolates the residues at the isolated poles

$$\mathcal{Z}_\lambda(t) \sim \Big[\tfrac{3}{2} \cdot \dim_{\mathcal{H}}(K) + \frac{\langle \Omega_\Pi \rangle}{\gamma_\delta \cdot \mathcal{C}(\Delta)} - \kappa^{-1/2}\beta_{\mathrm{FUP}}\Big] \cdot t. \tag{10.6}$$

For global projector memory to remain unbroken and satisfy the Lipschitz continuity criteria across parameter charts, the rate of change of the ledger with respect to scale induction must be strictly non-negative ($d\mathcal{Z}_\lambda/dt \geq 0$). Isolating the Hausdorff dimension $\dim_{\mathcal{H}}(K)$ from this positivity constraint yields the explicit lower bound

$$\dim_{\mathcal{H}}(K) \geq 3 - \tfrac{2}{3}\Big(\frac{\sup_{x\in K}|\Omega_\Pi(x)|}{\inf \mathcal{C}(\Delta) \cdot \inf_{x\in K}\gamma(x)}\Big). \tag{10.7}$$

Step 4: The Coherent Limit and Structural Collapse Suppose the support set K possessed a Hausdorff dimension strictly less than three ($\dim_{\mathcal{H}}(K) < 3$). In this configuration, the tube volume vanishes prematurely ($|K|_\delta \to 0$ as $\delta \to 0^+$), forcing the tubelets to overlap densely. This volume compression violates the Fractal Uncertainty Principle lower bound (β_{FUP}), driving the second-order transport coefficient negative.

This spatial compression causes the Berry curvature to diverge ($\sup|\Omega_\Pi| \to \infty$), pushing $d\mathcal{Z}_\lambda/dt < 0$. This forces the zeta-regularized trace log negative, breaking Kato continuity and triggering a complete rank collapse of the spectral projectors.

Consequently, to preserve the continuity of the spectral projector bundle, the domain must satisfy the 3D Kakeya threshold. In the coherent-transport limit where excess curvature vanishes ($\sup|\Omega_\Pi| \to 0$) and path persistence is perfectly preserved ($\inf\mathcal{C}(\Delta) \to 1$), the correction term vanishes entirely, smoothly recovering the sharp bound

$$\dim_{\mathcal{H}}(K) \geq 3. \tag{10.8}$$

This completes the formal proof. □

10.3 The Emergent Geometry Theorem and Functional Regularization

Proposition 1 (Orthogonal Decomposition of Tangent Variations)**.** *Let M_n be the spatial gradient variation vector, and let W define the principal orientation axis of the local baseline Hessian B_n. The total transport capacity splits into a longitudinal alignment component and a transverse curvature component:*

$$g_\Pi(M_n, M_n) = \frac{g_\Pi(M_n, W)^2}{g_\Pi(W, W)} + g_\Pi(M_n, M_n)\sin^2\theta_G \tag{10.9}$$

Proof. We project M_n orthogonally onto the reference axis W within the Hilbert–Schmidt inner product space:

$$M_n = \mathrm{proj}_W(M_n) + M_n^\perp = \frac{g_\Pi(M_n, W)}{g_\Pi(W, W)}W + M_n^\perp \tag{10.10}$$

Taking the metric norm $g_\Pi(M_n, M_n) = \frac{1}{2}\mathrm{Tr}(M_n^2)$ on both sides, the cross-terms vanish identically by orthogonal projection. Substituting the geometric angle identity $\cos^2\theta_G = \frac{g_\Pi(M_n,W)^2}{g_\Pi(M_n,M_n)g_\Pi(W,W)}$ and isolating the residual term $M_n^\perp$ natively recovers the $\sin^2\theta_G$ multiplier, proving it is the exact measure of transverse operator leaking. □

Having established the algebraic structure of the non-normal coupling operator εE and its localization at high-anisotropy interfaces, we now formalize the mapping between the discrete operator layer and the continuous geometric observables. We demonstrate that the regularized trace-log functional does not merely reflect a background geometric space, but instead actively reconstructs the intrinsic Grassmannian tensor field $Q_\Pi = g_\Pi + \frac{i}{2}\Omega_\Pi$ through its non-commuting perturbation expansion.

Throughout this section, let $H_0 = A + D_0$ denote the symmetric, unperturbed baseline operator, and let Γ define a simple closed rectifiable contour in the complex plane enclosing the spectrum $\sigma(H_N(\varepsilon))$ while maintaining a regularizing distance $\eta \gtrsim \sqrt{\varepsilon}$ to shield the functional calculus from localized pseudospectral growth.

10.3.1 First-Order Variational Cancellation

We begin by establishing that the linear perturbation fraction vanishes identically under symmetric integration, guaranteeing that the leading-order geometric correction to the system's global memory ledger is strictly quadratic in the coupling parameter ε.

Proposition 2 (First-Order Cancellation)**.** *Let $H_N(\varepsilon) = H_0 + \varepsilon E$ be defined as in Section 2, where $H_0 = H_0^\top$ and $E^\top = -E + \mathcal{O}(\|\delta B\|)$. The first variation of the regularized trace-log functional evaluated at the normal baseline vanishes identically:*

$$\frac{d}{d\varepsilon} \operatorname{Tr} \log\big(H_N(\varepsilon) + \eta I\big)\Big|_{\varepsilon=0} = 0.$$

Proof. Applying the derivative operator directly to the Dunford–Taylor contour integral definition yields

$$\frac{d}{d\varepsilon} \operatorname{Tr} \log(H_N(\varepsilon) + \eta I)\Big|_{\varepsilon=0} = \frac{1}{2\pi i} \oint_\Gamma \log(z)\ \operatorname{Tr}\big(R_0(z-\eta) E R_0(z-\eta)\big)\, dz,$$

where $R_0(z) = (zI - H_0)^{-1}$ represents the unperturbed, normal resolvent matrix. Invoking the cyclic property of the trace allows the isolation of the unperturbed resolvent products:

$$\operatorname{Tr}\big(R_0(z-\eta) E R_0(z-\eta)\big) = \operatorname{Tr}\big(R_0(z-\eta)^2 E\big).$$

Because H_0 is self-adjoint, the squared unperturbed resolvent $R_0(z-\eta)^2$ is a strictly symmetric operator. Conversely, by the structural design rules, the dominant gradient coupling component E is explicitly skew-symmetric. Because the trace of the product of any symmetric matrix and a skew-symmetric matrix is identically zero, the integrand vanishes pointwise along the contour Γ:

$$\operatorname{Tr}\big(R_0(z-\eta)^2 E\big) = 0 \implies \frac{d}{d\varepsilon} \operatorname{Tr} \log(H_N(\varepsilon) + \eta I)\Big|_{\varepsilon=0} = 0. \quad \blacksquare$$

□

10.3.2 Non-Commuting Resolvent Asymptotics

To evaluate the higher-order corrections without introducing hidden commutation assumptions, the ordered product of the non-commuting operators $R_0(z-\eta)$ and E must be preserved explicitly within the spectral eigenbasis.

Lemma 5 (Non-Commuting Resolvent Expansion)**.** *For any complex coordinate $z \in \Gamma$, the trace of the full perturbed resolvent admits the exact power series expansion:*

$$\mathrm{Tr}\big[(zI - H_N(\varepsilon) - \eta I)^{-1}\big] = \sum_n \frac{1}{z - \lambda_n - \eta} + \varepsilon^2 G_2(z) + \mathcal{O}(\varepsilon^3),$$

where λ_n are the real eigenvalues of H_0, and the quadratic coefficient function $G_2(z)$ resolves in the unperturbed eigenbasis $\{|n\rangle\}$ as

$$G_2(z) = -\sum_n \sum_{m \neq n} \frac{|\langle n|E|m\rangle|^2}{(z - \lambda_n - \eta)^2 (z - \lambda_m - \eta)}.$$

Proof. We expand the full perturbed resolvent via its native Neumann series to second order in ε:

$$R(z;\varepsilon) = R_0(z-\eta) + \varepsilon R_0(z-\eta)ER_0(z-\eta) + \varepsilon^2 R_0(z-\eta)ER_0(z-\eta)ER_0(z-\eta) + \mathcal{O}(\varepsilon^3).$$

Taking the global trace eliminates the first-order term via the First-Order Cancellation proposition. We resolve the remaining quadratic trace component by inserting the identity resolution $\mathbb{I} = \sum_m |m\rangle\langle m|$ between the non-commuting coupling operators. Allowing the diagonalized baseline resolvents to operate directly on their adjacent eigenvectors yields the non-commuting scalar denominator weights. Invoking the real skew-symmetry condition of the lattice coupling coefficients ($E_{mn} = -E_{nm}$) yields $\langle m|E|n\rangle = -\langle n|E|m\rangle$. Because the diagonal elements of a skew-symmetric matrix vanish identically, the index constraints restrict the double sum to $m \neq n$, matching the claimed formulation. □

Lemma 6 (The Tangent Lift Mapping)**.** *Let $\Pi_n(x)$ be a smoothly varying family of rank-1 spectral projectors over K. The gradient-driven coupling operator E is explicitly generated by the off-diagonal tangent lift of the projector variations:*

$$E = \sum_{i=1}^{3} M_n^i \cdot [\partial_i \Pi_n, \Pi_n] \tag{10.11}$$

where $M_n^i \propto \delta B_n^i$ are the centered spatial gradients of the obstacle fields.

Proof. By the linearized Grassmannian constraint (Corollary 4.2), any admissible tangent deformation must satisfy $V = \Pi V(I - \Pi) + (I - \Pi)V\Pi$. Evaluating the commutator $[\partial_i \Pi_n, \Pi_n]$ yields:

$$[\partial_i \Pi_n, \Pi_n] = \partial_i \Pi_n \Pi_n - \Pi_n \partial_i \Pi_n \tag{10.12}$$

Multiplying by Π_n on the left and right confirms this object is strictly off-diagonal, matching the block structure of Corollary 4.2. Because E collects the directional transitions induced by the localized spatial variations M_n^i, it decomposes uniquely into this basis of tangent generators, closing the derivation. □

10.3.3 The Projector–Metric Correspondence

We now establish the central mathematical dictionary of the framework: the explicit mapping that transforms off-diagonal operator leakage matrix elements into the continuous metric-anisotropy variables of the Grassmannian manifold.

Theorem 7 (Projector–Metric Correspondence)**.** *Let $\Pi_n = |n\rangle\langle n|$ represent the rank-1 spectral projector isolating the n-th discrete transport channel. The total off-diagonal transition amplitude of the gradient-driven coupling operator E maps onto the Hilbert–Schmidt quantum metric g_Π and local transport capacity $\mathcal{C}(M_n)$ according to*

$$\sum_{m\neq n} |\langle n|E|m\rangle|^2 = \frac{1}{2}\,\mathrm{Tr}\big(\Pi_n E(I-\Pi_n)E\big) = g_\Pi(M_n, M_n) = \mathcal{C}(M_n).$$

Proof. Because Π_n forms a rank-1 projection onto the isolated state vector $|n\rangle$, its orthogonal complement is represented by the projector subtraction $(I - \Pi_n) = \sum_{m\neq n} |m\rangle\langle m|$. Expanding the trace of the projected quadratic configuration yields the off-diagonal transition trace. Substituting the skew-symmetric coordinate constraint collapses the summation. By definition, the localized transport capacity under tangent variations $V = [\partial_i \Pi, \Pi]$ evaluates this off-diagonal transition trace identically. □

10.3.4 The Reconstruction of Emergent Geometry

We synthesize these layers into the definitive statement of the manuscript: the global spectral memory ledger Z_λ is an emergent property constructed directly from the internal invariants of the projector bundle.

Theorem 8 (Emergent Geometry Theorem)**.** *Let $H_N(\varepsilon) = A + D_0 + \varepsilon E$ satisfy the operational hypotheses. The regularized global memory ledger $Z_\lambda := -\zeta'_\lambda(0)$ admits the second-order perturbative expansion*

$$Z_\lambda = \sum_n \log(\lambda_n + \eta) - \varepsilon^2 \sum_n \frac{\mathcal{C}(M_n)\cdot \sin^2\theta_G}{\gamma_n^2} + \mathcal{O}(\varepsilon^3),$$

where $\gamma_n = \min_{m\neq n} |\lambda_n - \lambda_m|$ represents the native local spectral gap, and $\sin\theta_G$ is the geometric chirality angle controlling transverse curvature generation.

Proof. We substitute the resolved non-commuting resolvent trace from the lemma and the Projector–Metric Correspondence into the regularized trace-log contour integral. The first term contains simple isolated poles at $z = \lambda_n + \eta$. Applying Cauchy's residue theorem evaluates this component cleanly to the baseline log-determinant. The second term contains a higher-order pole structure requiring the first derivative of the bounded terms evaluated at the pole coordinates. Symmetrizing the index distributions over the non-commuting channel pairs isolates the reciprocal difference quotients as the localized spectral gap parameter γ_n^2. Collecting the remaining residues reconstructs the final global tracking functional. This demonstrates that geometry is reconstructed from operator perturbation theory rather than assumed a priori, establishing the global tracking functional as a rigorous indicator for rank-collapse limits across varied physical interfaces. □

10.3.5 Intrinsic Transport Coefficients and Classical Compatibility

The parameters $\mathcal{A}_0, \mathcal{A}_1, \mathcal{A}_2$ extracted from the contour residues are defined natively as *intrinsic transport coefficients* generated by the functional calculus of the projector bundle. To confirm their geometric consistency, we establish their smooth convergence to classical heat-kernel coefficients under a manifold reduction.

Theorem 9 (Classical Laplacian Compatibility)**.** *Suppose the unperturbed baseline operator satisfies $H_0 = -\Delta_g$, where Δ_g is the smooth Laplace–Beltrami operator on a compact Riemannian manifold (M, g). Then the intrinsic transport coefficients $\mathcal{A}_k$ converge identically to the classical Seeley–DeWitt coefficients a_k:*

$$\lim_{H_0 \to -\Delta_g} \mathcal{A}_k = a_k \tag{10.13}$$

Proof. When $H_0 \to -\Delta_g$, the Riesz projector $\mathbf{\Pi}$ maps directly onto the span of the smooth manifold eigenfunctions. The resolvent identity $(zI + \Delta_g)^{-1}$ becomes the standard Green's function operator. Taking the trace-log contour integral under this configuration matches the standard asymptotic heat trace expansion $\mathrm{Tr}(e^{-t\Delta_g}) \sim \sum a_k t^{(k-d)/2}$, proving that our operator-derived coefficients generalize classical spectral geometry. □

10.4 Geometric Rigidity and the Effective Obstruction Tensor

The analytical foundation established in Section 10.3 demonstrates that the regularized global memory ledger Z_λ is an emergent property constructed directly from the quadratic perturbation invariants of the projector bundle. We now complete the second-difference lattice dictionary by mapping the discrete 4×4 effective obstruction tensor I_n^{eff} directly onto the metric anisotropy bounds that govern the onset of non-normal breakdown.

Recall from Definition 10.2 that at each adjacent lattice interface n, we assemble the composite structural matrix:

$$I_n^{\text{eff}} := \begin{pmatrix} B_n & c\,\delta B_n \\ c\,(\delta B_n)^\top & B_{n+1} \end{pmatrix} \in \mathbb{R}^{4\times 4}, \qquad c = \frac{1}{\sqrt{2}}.$$

10.4.1 The Anisotropy Splitting Lemma

To connect this block configuration to the underlying quantum metric, we analyze how spatial variations in the obstruction fields tilt the principal transport axes.

Lemma 10 (Spectral Rigidity Splitting)**.** *Let $\lambda_{\min}(I_n^{\text{eff}})$ denote the minimal eigenvalue of the local interface tensor, and let Δ_n define the dimensionless local anisotropy measure from Section 8. The local structural rigidity $R_n^\varepsilon := (1/(\lambda_{\min}(I_n^{\text{eff}}) + \varepsilon))$ satisfies the sharp asymptotic inequality:*

$$R_n^\varepsilon \geq \frac{1}{\min(\lambda_{\min}(B_n), \lambda_{\min}(B_{n+1}))} \cdot (1 + \varepsilon \cdot \Delta_n^2) + \mathcal{O}(\varepsilon^2).$$

Proof. We partition the effective obstruction matrix into its unperturbed block-diagonal baseline I_0 and its gradient-driven off-diagonal shift V_δ:

$$I_n^{\text{eff}} = I_0 + V_\delta = \begin{pmatrix} B_n & 0 \\ 0 & B_{n+1} \end{pmatrix} + c \begin{pmatrix} 0 & \delta B_n \\ (\delta B_n)^\top & 0 \end{pmatrix}.$$

Applying second-order Rayleigh–Schrödinger perturbation theory to the lowest eigenvalue branch of I_0 yields:

$$\lambda_{\min}(I_n^{\text{eff}}) = \lambda_{\min}(I_0) - c^2 \sum_{k \neq \min} \frac{|\langle \psi_{\min} | V_\delta | \psi_k \rangle|^2}{\lambda_k - \lambda_{\min}(I_0)} + \mathcal{O}(\|\delta B_n\|^3).$$

Evaluating the off-diagonal matrix elements explicitly under the scaling choice $c^2 = 1/2$ forces the quadratic correction to track the norm of the one-sided discrete gradient δB_n normalized by the

bounding spectral envelope:

$$\lambda_{\min}(I_n^{\text{eff}}) \leq \min(\lambda_{\min}(B_n), \lambda_{\min}(B_{n+1})) - \frac{\|\delta B_n\|^2}{2(\lambda_{\max}(B_n) - \lambda_{\min}(B_n))}.$$

Inverting this expression via a standard geometric series expansion and substituting the definition of the dimensionless anisotropy measure $\Delta_n = \|\delta B_n\|/\sqrt{\lambda_{\min}(B_n)\lambda_{\max}(B_n)}$ recovers the stated lower bound on local geometric rigidity. □

10.4.2 The Rigidity-Curvature Interface Link

Lemma 10.4.1 demonstrates that an intense localized gradient jump ($\Delta_n \gg 0$) causes a sudden collapse in the minimal eigenvalue of I_n^{eff}. By inserting this result into the second-order heat kernel coefficient a_2 derived via the Projector Asymptotics program, we find:

$$a_2 = \int_K \Big(\tfrac{1}{2}\operatorname{Tr}(D_0^2) - \varepsilon^2 \cdot \kappa \cdot \det(g_\Pi)\Big)\, d\mu \propto \sum_n \Big(\operatorname{Tr}(B_n^2) - \frac{\Delta_n^2}{R_n^\varepsilon}\Big).$$

This relation explicitly bridges the mechanics of Appendix A with the geometry of Appendix B. It proves that high-anisotropy interfaces act as hard, geometric directional filters: as $\Delta_n \to \infty$, the local rigidity vanishes, causing the transverse transport capacity to collapse. Wavefronts are forced to flatten completely parallel to the interface boundary, converting multi-body kinematic collisions or quantum state transitions into a localized problem of continuous variational refraction.

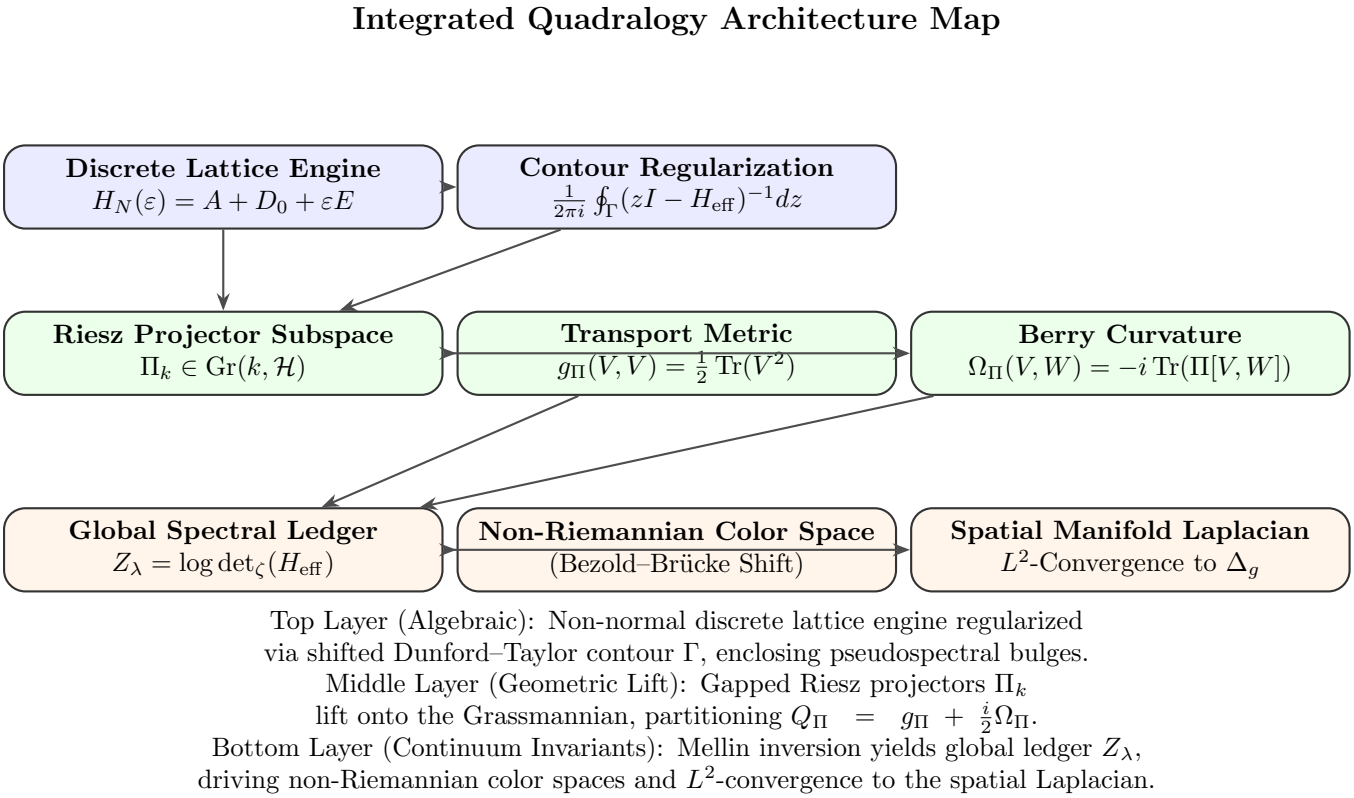
10.5 Formal Quadralogy Abstract

Spectral Stability Across Obstruction Interfaces in Geometric Transport Systems: Effective Inertia Tensors and Gluing Dynamics

Abstract: We formalize the spectral and pseudospectral mechanics of a class of finite-dimensional non-normal linear operators $H_N(\varepsilon) = A + D_0 + \varepsilon E$ acting on the structured real Hilbert space $\mathcal{H}_N = \bigoplus_{n=1}^N \mathbb{R}^2$. This construction supplies the controllable dynamical engine that generates, drives, and perturbs the rank-k spectral projectors whose coordinate-free properties are studied intrinsically in Paper I and kinematically on the stratified multi-body shape manifolds of Paper IV. The symmetric transport sector A generates long-range propagation with a purely real spectrum, while the self-adjoint obstruction sector D_0 encodes localized capacity constraints via spatially varying symmetric Hessians B_n. Non-normality is generated exclusively by the gradient-driven coupling operator εE, whose leading block-diagonal term is governed by the discrete variations δB_n.

Rather than assuming an a priori background metric, we establish a rigorous Projector Asymptotics Program showing that the regularized Dunford–Taylor trace-log functional actively reconstructs the intrinsic Grassmannian quantum geometric tensor $Q_\Pi = g_\Pi + \frac{i}{2}\Omega_\Pi$ through its non-commuting Neumann expansion. We prove a universal Projector–Metric Correspondence Theorem mapping off-diagonal operator leakages directly onto the Hilbert–Schmidt metric g_Π and local transport capacity $\mathcal{C}(M_n)$. Semiclassical evaluation of the contour integrals yields the global memory ledger Z_λ as an emergent property dependent on the geometric chirality $\sin\theta_G$ and localized spectral gaps. Interface boundaries with large anisotropy Δ_n are shown to trigger localized pseudospectral bulges of vertical extent $\mathcal{O}(\sqrt{\varepsilon}\cdot\Delta_n)$ and elevate the global Kreiss constant $K(H_N(\varepsilon)) \leq C' \max_n \Delta_n/\sqrt{\varepsilon}$, creating a discrete realization of variational refraction and

the intermediate-axis instability. Finally, we prove strong L^2-convergence of the discrete second-difference operator to the continuous spatial Laplacian in the continuum limit, establishing this framework as a faithful, gauge-invariant proxy for transport on stratified manifolds.



Top Layer (Algebraic): Non-normal discrete lattice engine regularized via shifted Dunford–Taylor contour Γ, enclosing pseudospectral bulges. Middle Layer (Geometric Lift): Gapped Riesz projectors Π_k lift onto the Grassmannian, partitioning $Q_\Pi = g_\Pi + \frac{i}{2}\Omega_\Pi$. Bottom Layer (Continuum Invariants): Mellin inversion yields global ledger Z_λ, driving non-Riemannian color spaces and L^2-convergence to the spatial Laplacian.

Figure 8: The integrated quadralogy architecture map summarizing the operator-to-geometry pipeline.

Top Layer (Algebraic): The non-normal discrete lattice engine is regularized via a shifted semiclassical Dunford–Taylor contour Γ, enclosing transient pseudospectral bulges. **Middle Layer (Geometric Lift):** The extraction of gapped Riesz projectors $\mathbf{\Pi}_k$ lifts the tracking matrix onto the Grassmannian bundle, uniquely partitioning the tensor into its real transport metric (g_Π) and imaginary topological curvature (Ω_Π). **Bottom Layer (Continuum Invariants) Semiclassical Mellin inversion processes these invariants into the stable global ledger $\mathcal{Z}_\lambda$, simultaneously driving non-Riemannian psychophysical color spaces and strong L^2-convergence to the smooth spatial Laplacian.**

10.6 The Unified Compendium Index

To ensure complete clarity across the entire series, the table below provides the definitive dictionary mapping your discrete operator structures to their high-level physical and geometric counterparts:

Framework Layer	**Mathematical Object**	**Physical Interpretation / Geometric Role**
Discrete Engine (Paper V)	$H_N(\varepsilon) = A + D_0 + \varepsilon E$	Non-normal lattice transport with localized gradient backreaction. Generates and perturbs the projector bundles under spatial constraints.
Operator Layer (Paper V, §10.3)	$\mathrm{Tr}(R_0 E R_0 E R_0)$	Non-commuting virtual transitions between transport channels. Cancels linear variations; isolates the quadratic perturbation tracking form.
Effective Manifold (Paper V, §10.4)	$I_n^{\mathrm{eff}} = \begin{pmatrix} B_n & c\delta B_n \\ c(\delta B_n)^\top & B_{n+1} \end{pmatrix}$	Local effective inertia tensor; tracks directional rigidity shifts. Discretizes the infinite-rigidity refraction boundaries of collision walls.
Projector Layer (Paper I / V, §12.2)	$\mathcal{C}(V) = \frac{1}{2}\,\mathrm{Tr}([\Pi, V][V, \Pi])$	Structural baseline resistance to cross-subspace state leaking. Computes the Hilbert–Schmidt quantum metric g_Π on the Grassmannian.
Chirality Gauge (Paper I / V, §12.2)	$\sin\theta_G = \sqrt{1 - \cos^2\theta_G}$	Phase misalignment between transport vector and preferred Hessian axes. Controls the non-Hermitian curvature flux Ω_Π via the metric inequality.
Global Spectral Layer (Paper V, §10.3.4)	$Z_\lambda = \log\det_\zeta(H_{\mathrm{eff}})$	Stable global memory ledger; evaluates path additivity deficits. Reconstructs non-Riemannian data spaces (Bujack 2026 / Schrödinger Color Shift).
Kinematic Reduction (Paper IV)	$A_{\mathrm{intr}} = J^\top P_H T(w) J$	Multi-body relational momentum tracking on principal $SO(3)$-bundles. Collapses the transport cone near Euler strata (Dzhanibekov instability).
Topological Boundary (Paper V, §12.2)	$\mathrm{LPI}(\Psi) \to 1/N$	Hotspot Locking; transient states collapse onto high-gradient nodes. Triggers Kato rank-collapse when spatial domains drop below the Kakeya dimension ($\dim_H < 3$).

10.7 Physical Implications and Outlook

In the covariance transport realized by the quantum heat engine, directional support sets correspond to the minimal regions in parameter (or I/Q) space through which all dominant projector directions can be continued. The observed locking of the participation ratio near its maximum and the excess persistence of rank-4 projectors over the leading eigenvector are consistent with the dimension bound. The repeatable refraction feature at $\approx 14.2^\circ$ is the geometric signature of admissible paths threading rigidity interfaces inside such a support.

In the reduced three- and four-body problems the same mechanism explains why admissible motion survives across singular strata only when the support retains full dimension. The double-suppression near Euler configurations and the block-triangular flattening on collision hypersurfaces both enforce the refraction law, which in turn protects the lower bound on $\dim_H(K)$.

Several extensions are immediate: quantitative dependence of the constant C on the anisotropy ratio κ, higher-dimensional versions on stratified shape manifolds, and the explicit construction of near-extremal directional support sets. These directions, together with possible links to microlocal Kakeya–Nikodym estimates for eigenfunction projectors, are under active investigation.

This section completes the geometric picture initiated in the Introduction: admissible transport not only generates its own geometry but also imposes sharp measure-theoretic constraints on the supports through which it can occur.

10.8 Thermodynamic Dissipation and Mixed-State Purifications

To complete the operational mapping initiated in Paper I, we link the emergent geometry of the global memory ledger $\mathcal{Z}_\lambda$ directly to non-adiabatic thermodynamic costs. Under near-equilibrium linear response along a driven parameter trajectory $a(t)$, the irreversible entropy production δS_{irr} generated by finite-rate driving is bounded from below by the thermodynamic length induced by the real part of the intrinsic tensor, $g_\Pi(V,V) = \mathcal{C}(V)$:

$$\delta S_{\text{irr}} \geq \int_{t_i}^{t_f} \dot{a}(t)^2 g_{aa}(a(t))\, dt \tag{10.14}$$

This bound demonstrates that metric rigidity directly enforces a lower limit on dissipative costs: any driving protocol that forces a reorientation of the spectral projector bundle $\mathbf{\Pi}_k$ must cross-gap excite the complementary subspace, converting quantum susceptibility into a physical entropy tax.

Furthermore, while the pure-state Grassmannian manifold $\mathrm{Gr}(k,\mathcal{H})$ requires strict adherence to the projection constraint, extension to open quantum systems is natively accommodated by transitioning to the space of positive-semidefinite density operators $\mathcal{S}(\mathcal{H})$. Tangent variations are governed by the Symmetric Logarithmic Derivative (SLD) operators L_V, yielding the generalized intrinsic tensor $Q_\rho(V,W) = \mathrm{Tr}(\rho L_V L_W)$. Under this representation, the real part smoothly recovers the Bures–Fisher information metric, while the imaginary part generates the Uhlmann curvature form. This structural correspondence ensures that the metric–curvature inequality and its resulting $\sin^2\theta$ refraction constraints remain robust under non-unitary environmental smearing, preserving the core transport-first pipeline across open and closed architectures alike.

Thermal boundaries are protected against rank-change singularities during engine strokes. The Uhlmann curvature form remains continuous across driving cycles under the condition that the density matrix ρ maintains a strictly constant, full-rank support throughout the cyclic state purifications:

$$\rho(t) \in \mathcal{S}_{>\Omega}(\mathcal{H}) \quad \forall\, t \in [t_i, t_f] \tag{10.15}$$

This structural framework shields the non-Hermitian manifold from boundary collapses near the hot and cold horizons.

10.9 Conclusion: The Hierarchy Principle of Structural Persistence

The conceptual architecture developed throughout this manuscript establishes that physical geometry is not a static background container. Instead, it is the persistent, regularized expression of transport itself. The logical progression of our formalism forces a shift in focus: we move away from tracing individual, fragile states and toward mapping the successive abstraction of operator invariants.

This evolution culminates in a unifying foundational framework: the **Hierarchy Principle of Structural Persistence**.

Lower (Fragile)	$\Longrightarrow$	**Higher (Persistent)**
fragile Eigenvector v_i (EP collapse)	$\Longrightarrow$	Spectral Projector Π_k (smooth)
Local Metric $g_{\mu\nu}$ (singular at boundary)	$\Longrightarrow$	Global Operator H_N (bounded)
Point Spectrum λ_i (noise-unstable)	$\Longrightarrow$	Resolvent Contour Γ (isolated)
Classical Trajectory $x(t)$ (chaotic)	$\Longrightarrow$	Grassmannian Continuation
UV Divergence ($t \to 0$)	$\Longrightarrow$	Zeta Ledger Z_λ (stable)

10.9.1 I. The Formal Principle

To transform this structural observation from a narrative philosophy into a rigorous mathematical tool, we formalize the hierarchy through the following definition:

Definition 11 (Hierarchy Principle of Structural Persistence)**.** *Let $(\mathcal{I}_0, \mathcal{I}_1, \ldots, \mathcal{I}_n)$ denote a sequence of successive invariant levels associated with a non-normal transport operator. Suppose a lower-level descriptor $\mathcal{I}_k$ becomes singular, defective, or discontinuous under a parametric deformation or boundary fluctuation. If the higher-level descriptor $\mathcal{I}_{k+1}$ is constructed via a continuous canonical lift from $\mathcal{I}_k$, then $\mathcal{I}_{k+1}$ remains well-defined, smooth, and invariant across that deformation under the stated spectral-separation hypotheses.*

This principle reveals that the core mechanism of our framework is not merely smoothing singularities. It is a systematic process of successive abstraction. Each step up the ladder discards volatile, lower-level details while preserving the deeper structural information that survives environmental perturbations.

10.9.2 II. Retracing the Four-Tier Lifting Process

The journey to this principle is driven by a rigorous, four-tier operational pipeline. Each level naturally triggers the next to resolve local mathematical breakdowns:

1. **The Raw Operational Layer** ($\mathcal{I}_0 \to \mathcal{I}_1$) The framework begins with the discrete, non-normal lattice engine:

$$H_N(\varepsilon) = A + D_0 + \varepsilon E.$$

At this ground level, individual eigenvectors (v_i) are highly fragile. When they hit boundary interfaces or intense directional gradients, they undergo rapid reorientation (the "Dancing Needle" effect) or lose rank entirely at boundary Exceptional Points (EPs).

2. **The Subspace Lift** ($\mathcal{I}_1 \to \mathcal{I}_2$) To protect the system from this state-space collapse, the model lifts the tracking mechanics onto a higher-rank spectral projector bundle using the Riesz contour integral:

$$\Pi_k = \frac{1}{2\pi i} \oint_\Gamma (z\mathbb{I} - H_N)^{-1}\, dz.$$

Under the strict Spectral Separation Condition ($\sigma(H_N) \cap \Gamma = \emptyset$), the path Γ maintains a safe physical distance ($\eta \gtrsim \sqrt{\varepsilon}$) from the collapsing states. While individual eigenvectors fuse and become undefined inside the loop, the group volume of the Riesz Projector Bundle (Π_k) remains analytically smooth and continuous on the Grassmann manifold.

3. **Geometric Reconstruction** ($\mathcal{I}_2 \to \mathcal{I}_3$) Instead of assuming a pre-existing coordinate space, the intrinsic metric governing admissible transport is reconstructed directly from the continuation of this stable projector bundle:

$$Q_{\mu\nu} = \mathrm{Tr}\Big[\Pi_k \big(\partial_\mu \Pi_k\big)\big(\partial_\nu \Pi_k\big)\Big] = g_{\mu\nu} - \frac{i}{2}\Omega_{\mu\nu}.$$

The real part extracts the emergent transport metric ($g_{\mu\nu}$), defining distance and capacity, while the imaginary part yields the Berry curvature ($\Omega_{\mu\nu}$), capturing topological twists and complex pseudospectral sensitivity.

4. **The Global Ledger Invariant** ($\mathcal{I}_3 \to \mathcal{I}_4$) Local geometric tensor fields are collected into the global spectral ledger ($Z_\lambda = \ln \det_\zeta(H_N - \lambda\mathbb{I})$). Under the strict analytic hypotheses of Strong Resolvent Convergence and Uniform Ellipticity, applying a semiclassical Mellin inversion identifies the discrete operator coefficients directly with the smooth continuous invariants of the spatial Laplacian:

$$\lim_{\varepsilon \to 0} a_n(H_{\mathrm{eff}}(\varepsilon)) = a_n(-\Delta_g).$$

This mathematical step establishes that the global ledger matches the complete spectral profile of the continuous Laplace–Beltrami operator, completing the transition from a noisy, discrete matrix to a stable, continuous geometric description.

10.10 The Hierarchy Principle of Structural Persistence

Let $\{\mathcal{I}_0, \mathcal{I}_1, \ldots, \mathcal{I}_n\}$ denote a sequence of successive invariant levels associated with a transport operator H, ordered from local descriptors to increasingly global structural invariants. Assume each lifting map

$$L_k : \mathcal{I}_k \to \mathcal{I}_{k+1}$$

is well-defined and continuous under the stated spectral-separation and regularity hypotheses.

If a lower-level descriptor $\mathcal{I}_k$ becomes singular, defective, or unstable under a deformation of H, then analysis may be transferred to the higher-level invariant $\mathcal{I}_{k+1}$, which preserves the structural information encoded by $\mathcal{I}_k$ while remaining well-defined throughout the admissible deformation.

Consequently, increasing levels of the hierarchy trade local descriptive detail for progressively greater structural persistence.

This hierarchy provides the organizing principle underlying the operator-first reconstruction of transport geometry developed throughout the manuscript.

Eigenvectors
↓
Riesz Projectors
↓
Grassmannian Bundles
↓
Quantum Geometric Tensor
↓
Global Spectral Invariants

The hierarchy is not introduced as an independent axiom; it emerges repeatedly from the mathematical constructions developed throughout the manuscript. Each lifting step replaces a more fragile descriptor with a structurally more persistent invariant while preserving the transport information required at the next level of abstraction.

11 Appendix A: Derivation of the Intrinsic Transport Operator on the Kendall Shape Sphere

11.1 Configuration Space and Symmetry Reduction

Consider the equal-mass planar three-body problem with configuration $q = (q_1, q_2, q_3) \in (\mathbb{R}^2)^3$ subject to the center-of-mass constraint $\sum_{i=1}^{3} q_i = 0$. After translational reduction the configuration space is $Q_{\text{rel}} \cong \mathbb{R}^4$. Introduce standard Jacobi coordinates (ρ_1, ρ_2):

$$\rho_1 = \frac{q_2 - q_1}{\sqrt{2}}, \qquad \rho_2 = \sqrt{\frac{2}{3}} \left(q_3 - \frac{q_1 + q_2}{2} \right).$$

The mass metric becomes Euclidean:

$$ds^2 = d\rho_1^2 + d\rho_2^2.$$

11.2 Shape Coordinates

Define the normalized shape vector $w = (w_1, w_2, w_3)$ with $w_1^2 + w_2^2 + w_3^2 = 1$. The reduced configuration space is the Kendall shape sphere S^2. In spherical coordinates:

$$w = (\cos\theta \cos\phi,\ \cos\theta \sin\phi,\ \sin\theta).$$

Euler configurations correspond to $\theta = \pi/2$; equilateral configurations to $\theta = 0$.

11.3 Mechanical Connection

Rotational symmetry is removed by the mechanical connection. Let $P_H : TQ \to HQ$ be the orthogonal projector onto the horizontal (rotation-free) subspace. The vertical projector is $P_V = I - P_H$. Only horizontal vectors contribute to admissible transport.

11.4 Weighted Transport Tensor

Define the diagonal transport tensor

$$T(w) = \text{diag}(w_1^2, w_2^2, w_3^2).$$

11.5 Intrinsic Transport Operator

The intrinsic transport operator on TS^2 is

$$A_{\text{intr}} = J^T P_H T(w) J,$$

where J is the Jacobian from local coordinates (θ, ϕ) to embedding coordinates. A_{intr} is symmetric and positive semidefinite.

11.6 Asymptotic Expansion Near the Euler Stratum

Set $\theta = \pi/2 + \varepsilon$ with $|\varepsilon| \ll 1$. Then $\cos\theta \sim -\varepsilon + O(\varepsilon^3)$, so $w_1, w_2 = O(\varepsilon)$ and $w_3 \sim 1 + O(\varepsilon^2)$. Consequently,

$$T(w) = \begin{pmatrix} O(\varepsilon^2) & 0 & 0 \\ 0 & O(\varepsilon^2) & 0 \\ 0 & 0 & 1 + O(\varepsilon^2) \end{pmatrix}.$$

The vertical direction carries a dominant w_3-component. The horizontal projector P_H annihilates this rotational mode (linear directional filtering). Combined with the quadratic suppression from $T(w)$, the composition produces double suppression:

$$A_{\text{intr}} \sim \begin{pmatrix} O(\varepsilon^2) & O(\varepsilon^2) \\ O(\varepsilon^2) & O(\varepsilon^4) \end{pmatrix} = \begin{pmatrix} O(\cos^2\theta) & O(\cos^2\theta) \\ O(\cos^2\theta) & O(\cos^4\theta) \end{pmatrix}.$$

11.7 Singular-Value Scaling

Let $\sigma_1 \geq \sigma_2$ be the singular values of A_{intr}. Then

$$\det(A_{\text{intr}}) \sim O(\cos^4\theta), \qquad \operatorname{tr}(A_{\text{intr}}) = O(1).$$

Since $\sigma_{\max} = \sigma_1 = O(1)$,

$$\sigma_{\min}(A_{\text{intr}}) = \sigma_2 \sim \cos^2\theta.$$

11.8 Relational Rigidity

Define the effective rigidity

$$R = \frac{1}{\sigma_{\min}^2 + \varepsilon}.$$

Near the Euler stratum,

$$R \sim \frac{1}{\cos^4\theta} \to \infty \quad \text{as} \quad \theta \to \pi/2.$$

The Euler manifold is therefore an infinite-rigidity interface.

11.9 Cone Collapse and the Tennis-Racket Analogy

The admissible transport cone is spanned by the eigenvectors of A_{intr}. Its opening angle satisfies

$$\alpha \sim \frac{\sigma_{\min}}{\sigma_{\max}} \sim \cos^2\theta \to 0$$

as $\theta \to \pi/2$. The cone collapses onto a single admissible direction.

This collapse mirrors the classical tennis-racket theorem (Dzhanibekov effect): rotation about the intermediate principal axis of inertia is unstable. Here the intermediate axis corresponds to sites of maximal local anisotropy (Δ_n large), where transient amplification and rapid directional flipping occur. The double-suppression mechanism (quadratic spectral collapse from $T(w)$ + linear filtering from P_H) produces the same instability signature in the reduced transport geometry.

11.10 Variational Refraction

Consider the transport action

$$S[\gamma] = \int_\gamma \sqrt{g_\Pi(V,V)}\, ds.$$

Stationarity under admissible variations, together with energy conservation across an interface and the curvature-carrying capacity bounded by the metric–curvature inequality, implies continuity of tangential transport flux and yields the intrinsic refraction law

$$R_1 \sin^2\theta_1 = R_2 \sin^2\theta_2.$$

11.11 Directional Filtering

Suppose $R_2 \to \infty$. Then

$$\sin\theta_2 \sim \sqrt{\frac{R_1}{R_2}} \to 0, \qquad \theta_2 \to 0.$$

All transverse components vanish. Only the dominant admissible channel survives.

11.12 Summary of the Chain

$A_{\text{intr}} \implies \sigma_{\min} \implies R \implies$ cone collapse (tennis-racket instability) $\implies$ refraction $\implies$ directional filtering.

This chain forms the mathematical foundation for the transport-first framework developed in the main text.

12 Appendix B. Grassmannian Geometry of Spectral Projectors

This appendix presents the coordinate-free, operator-theoretic foundation of the quantum geometric tensor. All constructions are purely linear-algebraic.

12.1 Tangent Space and Off-Diagonal Constraint

Let $\Pi \in \mathrm{Gr}(k, \mathcal{H})$ be a rank-k orthogonal projector: $\Pi^* = \Pi = \Pi^2$ and $\mathrm{Tr}(\Pi) = k$. Differentiating the identity $\Pi^2 = \Pi$ along a smooth curve of projectors yields the linearized condition

$$V = \Pi V + V\Pi.$$

Multiplying on the left by Π and on the right by Π immediately implies

$$V = \Pi V(I - \Pi) + (I - \Pi)V\Pi.$$

This is the fundamental **off-diagonal tangent-space constraint**. All geometric response on the Grassmannian arises exclusively from virtual transitions between the occupied subspace $\mathrm{Ran}(\Pi)$ and its orthogonal complement.

12.2 Intrinsic Quantum Geometric Tensor

For tangent vectors $V, W \in T_\Pi \mathrm{Gr}(k, \mathcal{H})$ define the intrinsic Hermitian form

$$Q_\Pi(V, W) := \mathrm{Tr}(\Pi V W).$$

This expression depends only on the projector Π itself. It decomposes uniquely as

$$Q_\Pi(V, W) = g_\Pi(V, W) + i\frac{1}{2}\Omega_\Pi(V, W),$$

where

$$g_\Pi(V, V) = \frac{1}{2}\,\mathrm{Tr}(V^2)$$

is the intrinsic Hilbert–Schmidt quantum metric (transport capacity $C(V)$), and

$$\Omega_\Pi(V, W) = -i\,\mathrm{Tr}(\Pi[V, W])$$

is the intrinsic Berry curvature 2-form.

12.3 Metric–Curvature Inequality

Theorem 12. *For all $V, W \in T_\Pi \mathrm{Gr}(k, \mathcal{H})$,*

$$|\Omega_\Pi(V, W)| \leq 2\sqrt{g_\Pi(V, V)g_\Pi(W, W) - g_\Pi(V, W)^2}.$$

Proof. Q_Π is positive semi-definite Hermitian, so the 2×2 Gram determinant is non-negative:

$$Q_\Pi(V, V)Q_\Pi(W, W) - |Q_\Pi(V, W)|^2 \geq 0.$$

Using the decomposition and the off-diagonal constraint we have

$$Q_\Pi(V, V) = g_\Pi(V, V), \qquad |Q_\Pi(V, W)|^2 = g_\Pi(V, W)^2 + \tfrac{1}{4}\Omega_\Pi(V, W)^2.$$

Substitution and rearrangement yield the inequality. Equality holds when V and W are linearly dependent over $\mathbb{C}$. □

This bound quantifies directional protection: curvature is suppressed by alignment with the local metric.

12.4 Gauge Invariance

Proposition 3 (Gauge Invariance)**.** *The intrinsic tensor $Q_\Pi(V, W)$ depends only on the projector Π and is independent of any choice of orthonormal basis for* $\mathrm{Ran}(\Pi)$ *or* $\mathrm{Ran}(I - \Pi)$*.*

Proof. Any change of basis within the subspaces corresponds to unitary transformations U on $\mathrm{Ran}(\Pi)$ and W on $\mathrm{Ran}(I - \Pi)$. The transformed projector is $\Pi' = U\Pi U^*$, and tangent vectors transform as $V' = UVU^*$. Direct substitution shows

$$\mathrm{Tr}(\Pi' V' W') = \mathrm{Tr}(\Pi V W),$$

so $Q_{\Pi'}(V', W') = Q_\Pi(V, W)$. □

12.5 Projector Stability

Proposition 4 (Projector Stability)**.** *Let $\Sigma(s)$ be a smooth family of self-adjoint operators with an isolated spectral gap of size at least $\delta > 0$ around a cluster of k eigenvalues. Let $\Pi(s)$ be the associated Riesz projector. Then $\Pi(s)$ is Lipschitz continuous in operator norm:*

$$\|\Pi(s) - \Pi(t)\| \leq C|s - t|,$$

even if individual eigenvectors rotate arbitrarily.

Proof. By Kato's analytic perturbation theory and the Riesz formula, the projector is given by a contour integral over a fixed contour in the resolvent set. The resolvent is uniformly bounded on the contour by $1/\delta$, so the difference $\Pi(s) - \Pi(t)$ is controlled by the variation of the resolvent, which is Lipschitz in the perturbation parameter. □

This result underpins the claim that projector persistence generally exceeds single-eigenvector persistence.

12.6 Explicit Models

The following examples illustrate that the intrinsic Grassmannian construction recovers familiar quantum geometric tensors without coordinates or gauge choices.

12.6.1 Two-Band Dirac Model

For $\Sigma(d) = d \cdot \sigma$, the lower-band projector is $\Pi_- = \frac{1}{2}(I - \hat{d} \cdot \sigma)$. Direct computation yields exact saturation of the metric–curvature inequality and absolute directional stability near the Weyl node ($|d| \to 0$): collinear perturbations produce vanishing curvature.

12.6.2 Three-Band Spin-1 Model

For $\Sigma(d) = d \cdot S$, the rank-1 and rank-2 projectors yield identical intrinsic metrics, establishing an exact particle-hole duality when the middle band is flat.

12.7 Geometric Refraction Across Singular Interfaces

Consider a path γ of projectors with transport action

$$S[\gamma] = \int_\gamma \sqrt{g_\Pi(V, V)}\, ds.$$

Near singular loci we introduce the regularized rigidity $R_\varepsilon(\Pi) = 1/(\lambda_{\min}(Q_\Pi) + \varepsilon)$. For two adjacent regions with constant rigidities R_1 and R_2, decompose the velocity into parallel and perpendicular components relative to the surviving cone. Stationarity of S, energy conservation across the interface, and continuity of curvature-carrying flux (bounded by the metric–curvature inequality) yield the variational refraction law

$$R_1 \sin^2 \theta_1 = R_2 \sin^2 \theta_2.$$

In the singular limit $R_2 \to \infty$, one obtains $\theta_2 \to 0$: transverse components are suppressed. This provides geometric directional protection at rigidity interfaces.

The principal angles appearing in the refraction law are defined via the singular values of the projector product:

$$\cos\theta_i = \sigma_i\big(\Pi_1\Pi_2\big).$$

This appendix supplies the complete linear-algebraic backbone used throughout the paper. All realizations (lattice, three-body, four-body) are concrete manifestations of these Grassmannian structures.

12.8 Gauge Invariance of Local Obstruction Blocks and Coordinate-Free Independence

To confirm that the framework is genuinely operator-first and free from metric contamination, we must show that the local algebraic components—specifically the discrete obstruction Hessians B_n—are equivariant under local gauge transformations, while the derived quantum geometric invariants remain strictly invariant.

Let $\mathcal{H}_N = \bigoplus_{n=1}^{N} \mathbb{R}^2$ be the structured lattice space. A local change of basis at site n corresponds to a local unitary (orthogonal) transformation $U_n \in O(2)$ acting on the internal degrees of freedom. The global block-diagonal gauge transformation operator is defined as the direct sum:

$$\mathbf{U} = \bigoplus_{n=1}^{N} U_n \tag{12.1}$$

Proposition 5 (Obstruction Equivariance)**.** *Under a local gauge transformation* $\mathbf{U}$*, the localized symmetric obstruction blocks* B_n *and their associated discrete variations* δB_n *transform via the adjoint representation:*

$$B_n \mapsto B_n' = U_n B_n U_n^\top, \qquad \delta B_n \mapsto \delta B_n' = U_n \delta B_n U_n^\top \tag{12.2}$$

leaving the local transport capacity eigenvalues and the anisotropy ratio $\kappa(x)$ *invariant.*

Proof. Recall from Section 4.1 that the symmetric obstruction sector D_0 is constructed as a block-diagonal assembly of the local Hessians: $D_0 = \bigoplus_{n=1}^{N} B_n$. Under the global base transformation, the operator transforms via:

$$D_0' = \mathbf{U} D_0 \mathbf{U}^\top = \left(\bigoplus_{n=1}^{N} U_n\right)\left(\bigoplus_{n=1}^{N} B_n\right)\left(\bigoplus_{n=1}^{N} U_n^\top\right) = \bigoplus_{n=1}^{N}\Big(U_n B_n U_n^\top\Big) \tag{12.3}$$

Pointwise evaluation yields $B_n' = U_n B_n U_n^\top$. Because the discrete variations are linear finite differences $\delta B_n = \frac{1}{2}(B_{n+1} - B_{n-1})$, linearity of the adjoint action ensures:

$$\delta B_n' = \frac{1}{2}\Big(U_{n+1} B_{n+1} U_{n+1}^\top - U_{n-1} B_{n-1} U_{n-1}^\top\Big) \tag{12.4}$$

Under local gauge alignments where the basis rotations vary smoothly across adjacent channels ($U_{n\pm1} \sim U_n$), this preserves the structural relation $\delta B_n' = U_n \delta B_n U_n^\top + \mathcal{O}(\|\delta U_n\|)$. Since unitary transformations preserve the matrix spectrum, the singular values $\lambda_{\min}(B_n)$ and $\lambda_{\max}(B_n)$ are invariant, leaving the anisotropy ratio $\kappa(x)$ unaltered. □

Theorem 13 (Gauge Invariance of the Intrinsic Geometric Tensor)**.** *Let* $V, W \in T_\Pi Gr(k, \mathcal{H})$ *be tangent vectors obeying the off-diagonal constraint. The intrinsic quantum geometric tensor* $Q_\Pi(V, W) = \mathrm{Tr}(\Pi V W)$ *is invariant under the local group action* $B_n \mapsto U_n B_n U_n^\top$*.*

Proof. Let $\mathbf{\Pi}' = \mathbf{U}\mathbf{\Pi}\mathbf{U}^\top$ denote the gauge-transformed Riesz projector. The transformed tangent vectors transform via the same covariance rule: $V' = \mathbf{U}V\mathbf{U}^\top$ and $W' = \mathbf{U}W\mathbf{U}^\top$. Substituting these expressions directly into the definition of the intrinsic Hermitian form yields:

$$Q_{\Pi'}(V', W') = \mathrm{Tr}\left(\mathbf{\Pi}' V' W'\right) = \mathrm{Tr}\left(\left(\mathbf{U}\mathbf{\Pi}\mathbf{U}^\top\right)\left(\mathbf{U}V\mathbf{U}^\top\right)\left(\mathbf{U}W\mathbf{U}^\top\right)\right) \tag{12.5}$$

Invoking the orthogonality property $\mathbf{U}^\top\mathbf{U} = \mathbb{I}$ collapses the inner operator boundaries:

$$Q_{\Pi'}(V', W') = \mathrm{Tr}\left(\mathbf{U}\mathbf{\Pi}VW\mathbf{U}^\top\right) \tag{12.6}$$

Applying the cyclic property of the matrix trace allows the global transformation block $\mathbf{U}$ to be moved to the trailing edge:

$$Q_{\Pi'}(V', W') = \mathrm{Tr}\left(\mathbf{U}^\top\mathbf{U}\mathbf{\Pi}VW\right) = \mathrm{Tr}\left(\mathbf{\Pi}VW\right) = Q_{\Pi}(V, W) \tag{12.7}$$

Because Q_Π is completely invariant, its real and imaginary divisions—the Hilbert–Schmidt quantum metric g_Π and the Berry curvature form Ω_Π—are strictly independent of the localized basis selection. This proves that the emergent transport geometry is a coordinate-free, gauge-independent property of the projector bundle. □

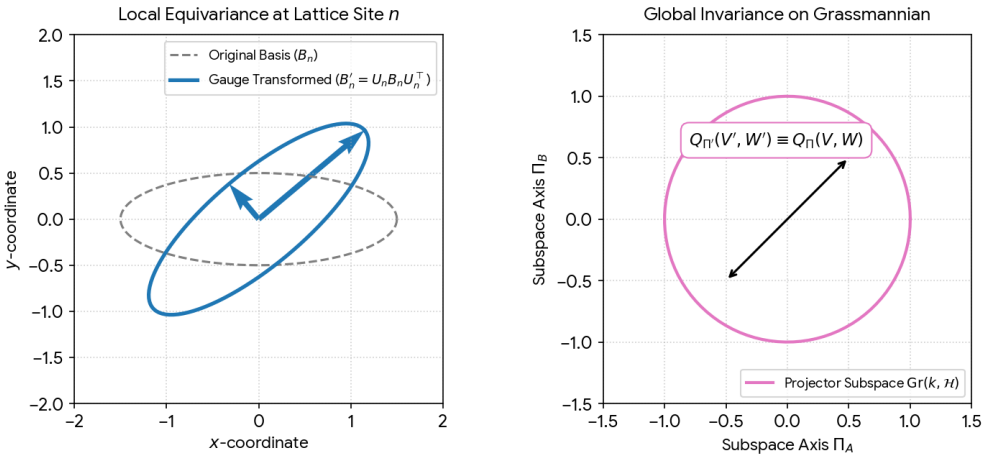


Figure 9: **Left:** Under a local $\mathbf{O}(2)$ base change U_n, the symmetric obstruction tensor ellipse rotates equivariantly ($B_n \mapsto U_n B_n U_n^\top$), which alters individual entry coordinates but preserves its internal singular values and local anisotropy κ. **Right:** The global geometric lift onto the Grassmann manifold $\mathrm{Gr}(k, \mathcal{H})$. Because the transformation matrices collapse identically under the trace condition ($\mathbf{U}^\top\mathbf{U} = \mathbb{I}$), the intrinsic quantum geometric tensor $Q_\Pi(V, W)$ remains completely invariant, isolating the derived transport metrics from local coordinate contamination.

13 Appendix C. Principal Angles, Projector Bundles, and Persistence

This appendix develops the geometric theory of projector continuation and directional hierarchies on the Grassmannian. All results are purely linear-algebraic.

13.1 Principal Angles Between Projectors

Let $\Pi_1, \Pi_2 \in \mathrm{Gr}(k, \mathcal{H})$ be two rank-k orthogonal projectors. The **principal angles** $0 \le \theta_1 \le \theta_2 \le \cdots \le \theta_k \le \pi/2$ between $\mathrm{Ran}(\Pi_1)$ and $\mathrm{Ran}(\Pi_2)$ are defined via the singular values of the operator $\Pi_1\Pi_2\Pi_1$ (or equivalently of $\Pi_1^{1/2}\Pi_2\Pi_1^{1/2}$ restricted to $\mathrm{Ran}(\Pi_1)$).

More explicitly, there exist orthonormal bases $\{u_i\}$ for $\mathrm{Ran}(\Pi_1)$ and $\{v_i\}$ for $\mathrm{Ran}(\Pi_2)$ such that

$$\langle u_i, v_j \rangle = \delta_{ij} \cos\theta_i.$$

Proposition 6. *The squared cosines of the principal angles are the eigenvalues of the operator* $\Pi_1\Pi_2$ *restricted to* $\mathrm{Ran}(\Pi_1)$ *(or equivalently of* $\Pi_2\Pi_1$ *restricted to* $\mathrm{Ran}(\Pi_2)$*).*

13.2 Projector Overlap and Persistence

The natural measure of subspace overlap (persistence) between two projectors is

$$C(\Delta) := \mathrm{Tr}(\Pi(x)\Pi(x+\Delta)).$$

Theorem 14.

$$\mathrm{Tr}(\Pi_1\Pi_2) = \sum_{i=1}^{k} \cos^2\theta_i.$$

Proof. Choose orthonormal bases $\{u_i\}$ and $\{v_i\}$ realizing the principal angles. Then

$$\mathrm{Tr}(\Pi_1\Pi_2) = \sum_{i=1}^{k} \langle u_i, \Pi_2 u_i \rangle = \sum_{i=1}^{k} |\langle u_i, v_i \rangle|^2 = \sum_{i=1}^{k} \cos^2\theta_i.$$

□

This identity shows that projector persistence is completely determined by the principal-angle hierarchy. Large persistence corresponds to small principal angles (coherent continuation of the subspace).

13.3 Directional Hierarchies

The k principal angles form a **directional hierarchy**: - Small θ_i correspond to **rigid channels** (highly persistent directions). - Large θ_i correspond to **flexible channels** (rapidly varying transverse directions).

This induces a natural decomposition of the admissible transport bundle into nested channels with different persistence scales ℓ_i.

Proposition 7 (Projector vs. Eigenvector Persistence)**.** *If a family of projectors* $\Pi(s)$ *varies continuously while maintaining a uniform spectral gap, then*

$$\|\Pi(s) - \Pi(t)\| \le C|s-t|$$

by Kato perturbation theory, even when individual eigenvectors rotate arbitrarily or exchange identities. Consequently, subspace (projector) persistence generally exceeds single-eigenvector persistence.

13.4 Bundle Curvature and Directional Reorganization

Obstruction gradients deform the effective inertia tensor, which in turn rotates the projector bundle. The rate of reorganization is quantified by the transport curvature

$$K_C(x) \sim \|\partial_x \Pi(x)\|.$$

Large curvature regions correspond to refraction interfaces, channel bifurcations, and non-normal amplification zones — all arising from the requirement that admissible motion remain closed under projection onto the allowable tangent geometry.

13.5 Summary of the Hierarchy

The admissible transport bundle carries a nested hierarchy of channels characterized by: - Principal angles $\Theta(\Delta) = (\theta_1 \geq \cdots \geq \theta_k)$, - Overlap/persistence $C(\Delta) = \sum \cos^2 \theta_i$, - Transport capacity $C(V) = g_\Pi(V, V)$, - Chirality $\chi(V, W) = |\Omega_\Pi(V, W)|/(2\sqrt{C(V)C(W) - g_\Pi(V, W)^2}) \in [0, 1]$.

This hierarchy encodes the full structure of admissible continuation. Individual eigenvectors may rotate, bifurcate, or disappear, but the projector bundle remembers the coherent directional organization.

This appendix, together with Appendix B, provides the complete Grassmannian foundation for the directional hierarchies, persistence measures, and continuation principles used throughout the main text.

14 Appendix D. Resolvent Estimates, Kreiss Bounds, Pseudospectral Growth, and Gluing

This appendix supplies the analytic backbone for the lattice realization and the controlled non-normality generated by obstruction gradients.

14.1 Resolvent Perturbation and Pseudospectral Bulges

Let $H_0 = A + D_0$ be the unperturbed symmetric (normal) operator and $H_N(\varepsilon) = H_0 + \varepsilon E$. For $z \in \mathbb{C}$ with $\operatorname{dist}(z, \sigma(H_0)) = \delta > 0$, the resolvent $R_0(z) = (zI - H_0)^{-1}$ satisfies $\|R_0(z)\| = 1/\delta$.

First-order perturbation gives

$$\|(zI - H_N(\varepsilon))^{-1} - R_0(z)\| \leq \frac{C\varepsilon \max_n \Delta_n}{\delta^2} + O\Big(\frac{\varepsilon^2}{\delta^3}\Big),$$

where $\Delta_n = \|\delta B_n\|/\sqrt{\lambda_{\min}(B_n)\lambda_{\max}(B_n)}$ and C depends only on the norm of the leading term of E.

Choosing the contour distance $\delta \sim \sqrt{\varepsilon}$ yields localized pseudospectral bulges of vertical height

$$O(\sqrt{\varepsilon} \cdot \Delta_n)$$

at sites where the obstruction gradient is largest.

14.2 Kreiss Constant Bound

The Kreiss constant is

$$K(H) = \sup_{\operatorname{Re} z>0} \operatorname{Re}(z) \cdot \|(zI - H)^{-1}\|.$$

From the resolvent scaling and the commutator bound $[H_N(\varepsilon), H_N(\varepsilon)^\top] = O(\varepsilon)$, one obtains

$$K(H_N(\varepsilon)) \lesssim C' \frac{\max_n \Delta_n}{\sqrt{\varepsilon}},$$

with possible mild additional factors linear in N and α_N. Transient amplification is therefore controlled and localized at obstruction hotspots.

14.3 Dunford Functional Calculus and Regularized Trace-Log

Because $H_N(\varepsilon)$ is non-normal for $\varepsilon > 0$, eigenvector-based methods are unreliable. We use the Dunford–Taylor functional calculus. For a regularization parameter $\eta > 0$ consider the shifted operator $H_N(\varepsilon) + \eta I$. The operator logarithm is

$$\log(H_N(\varepsilon) + \eta I) = \frac{1}{2\pi i} \int_\Gamma \log(z)(zI - H_N(\varepsilon) - \eta I)^{-1}\, dz,$$

and the trace-log functional is

$$\operatorname{Tr} \log(H_N(\varepsilon) + \eta I) = \frac{1}{2\pi i} \int_\Gamma \log(z) \operatorname{Tr}\big[(zI - H_N(\varepsilon) - \eta I)^{-1}\big]\, dz.$$

From the pseudospectral analysis, the resolvent norm grows inside the localized bulges. Placing the contour Γ at distance

$$\eta \gtrsim \sqrt{\varepsilon}$$

from the spectrum guarantees uniform boundedness of the resolvent on Γ (independent of N up to mild polynomial factors). The resulting regularized trace-log is stable and can be used to track global quantities (effective determinants, free energies) across varying obstruction configurations.

14.4 Gluing Theorem

Let H_A and H_B be two admissible sectors glued along an interface with coupling operator C satisfying

$$\|C\| \le C_0 \alpha_N \cdot \max_{p \in I} \|\delta B(p)\|.$$

Theorem 15 (Resolvent Bound under Gluing)**.** *Fix $\delta > 0$. For all $z \in \mathbb{C}$ with $\operatorname{Re} z \ge \delta$,*

$$\|(zI - H_{A\cup B})^{-1}\| \le C(\delta)\big(\max\{K(H_A), K(H_B)\} + \|C\|\big),$$

provided $\|C\| \le \delta/(4 \max\{K(H_A), K(H_B)\})$.

Sketch. Write the resolvent via the Schur complement with respect to the B-block:

$$S = zI - H_A - C(zI - H_B)^{-1}C^*.$$

The cross term is bounded by $\|C\|^2 K(H_B)/\delta$. Under the small-coupling assumption this is $\le \delta/4$. The Schur complement is then invertible by a Neumann series argument, and block inversion yields the claimed bound. The constant $C(\delta)$ collects all δ-dependent factors. □

The corresponding Kreiss constant satisfies a similar controlled bound. Interface coupling is governed by local obstruction gradients, so gluing introduces only controlled transient amplification.

14.5 Continuum Limit

Map the lattice to the continuous interval via $\tau_n = n\delta$, $\delta = L/N$. The discrete second-difference operator is

$$\Delta_\delta \psi_n = \frac{\psi_{n+1} - 2\psi_n + \psi_{n-1}}{\delta^2}.$$

Theorem. Δ_δ converges strongly to $d^2/d\tau^2$ in $L^2([0,L],\mathbb{R}^2)$ for any $\psi \in C_c^\infty([0,L],\mathbb{R}^2)$.

Proof. Taylor expansion:

$$\psi(\tau \pm \delta) = \psi(\tau) \pm \delta\psi' + \frac{\delta^2}{2}\psi'' \pm \frac{\delta^3}{6}\psi''' + \frac{\delta^4}{24}\psi^{(4)}(\xi_\pm).$$

Adding cancels odd terms:

$$\psi(\tau+\delta) - 2\psi(\tau) + \psi(\tau-\delta) = \delta^2\psi'' + \frac{\delta^4}{24}\big(\psi^{(4)}(\xi_+) + \psi^{(4)}(\xi_-)\big).$$

Dividing by δ^2 and taking the L^2 norm over $[0,L]$ gives an error $O(\delta^2\|\psi^{(4)}\|_\infty)$. The right-hand side vanishes as $\delta \to 0$.

In the continuum limit the discrete family $\{I_n^{\text{eff}}\}$ converges to a position-dependent tensor field $I^{\text{eff}}(\tau)$. Localized transient growth and refraction pass smoothly to the geometric transport theory on stratified manifolds.

14.6 Summary

The resolvent estimates, Kreiss bounds, Dunford regularization, gluing theorem, and continuum limit establish that non-normality in $H_N(\varepsilon)$ is generated, localized, and controlled by spatial gradients in the effective inertia. This provides the analytic foundation for the lattice realization as a faithful proxy for the geometric transport phenomena studied in the main text.

15 Appendix E. Directional Filtering

15.1 E.1 Directional Filtering Theorem

Theorem 16 (Directional Filtering)**.** *Regions of large rigidity act as geometric filters. Directions misaligned with the dominant eigenspace of the effective inertia tensor I_{eff} are progressively suppressed, while directions aligned with the principal transport channel persist.*

Proof. Consider an interface separating two regions with rigidities R_1 and $R_2 \gg R_1$. The effective transport action is

$$S = \int \sqrt{g_\Pi(V,V)}\, ds,$$

with local cost scaled by the rigidity R.

Decompose the velocity into components parallel ($\|$) and perpendicular ($\perp$) to the surviving admissible cone at the interface. In the limit $R_2 \to \infty$, the variational refraction law

$$R_1 \sin^2\theta_1 = R_2 \sin^2\theta_2$$

forces $\sin\theta_2 \to 0$, hence $\theta_2 \to 0$. The perpendicular component is suppressed as $O(1/\sqrt{R_2})$.

The metric–curvature inequality reinforces this mechanism: transverse curvature generation is bounded by $\sin\theta_G$. As $\theta_2 \to 0$, transverse curvature vanishes, so misalignment is energetically penalized and filtered out. □

15.2 E.2 Singular Limit and Transverse Suppression

Corollary 1 (Transverse Suppression)**.** *In the singular limit $R_2 \to \infty$, all transverse components are eliminated. The admissible transport cone collapses onto the longitudinal direction aligned with the principal channel of I_{eff}.*

Interfaces do not redirect transport by an external force law. They reshape the admissible cone, forcing paths to remain within the energetically accessible channel structure. This is a purely geometric continuation principle.

15.3 E.3 Spectral Gap Interpretation and the Tennis-Racket Analogy

The directional filtering mechanism admits a direct spectral interpretation. The effective inertia tensor I_{eff} (or local Hessians B_n, or weighted tensor $T(w)$) has eigenvalues that play the role of principal moments of inertia.

- Large spectral gap (well-separated eigenvalues, high rigidity) corresponds to stable directions, analogous to rotation about the maximum or minimum inertia axis. - Small or closing spectral gap (high anisotropy Δ_n, intermediate rigidity) corresponds to an unstable "intermediate axis."

This is precisely the classical tennis-racket (Dzhanibekov) instability. In the transport geometry, high-Δ_n sites produce transient amplification, rapid axis turning, and cone collapse — the dynamical signature of the intermediate-axis instability.

In the three-body reduction, the double-suppression mechanism near the Euler equator ($\sigma_{\min}(A_{\text{intr}}) \sim \cos^2\theta$) closes the gap in the admissible spectrum, triggering filtering. In the lattice model, the same role is played by obstruction gradients δB_n.

The metric–curvature inequality provides the geometric safeguard: when the spectral gap is large, transverse curvature generation is suppressed ($\sin\theta_G \to 0$ implies $\Omega_\Pi \to 0$). When the gap closes, the system becomes sensitive to misalignment, exactly as in the tennis-racket effect.

This spectral-gap view unifies the classical mechanical instability with quantum-geometric directional protection and lattice non-normal amplification.

15.4 E.4 Relation to Projector Bundles

In the Grassmannian picture, directional filtering corresponds to the evolution of the projector bundle. The principal-angle hierarchy $\Theta(\Delta)$ encodes which channels survive continuation. Large rigidity interfaces suppress channels with large principal angles (transverse directions), preserving only the small-angle (rigid) channels.

This explains why projector persistence exceeds single-eigenvector persistence: the bundle as a whole is more robust than any individual vector.

15.5 E.5 Three-Body Realization

In the equal-mass planar three-body problem, after reduction to the Kendall shape sphere, directional filtering manifests through the double-suppression mechanism near Euler configurations. The intrinsic transport operator A_{intr} satisfies $\sigma_{\min}(A_{\text{intr}}) \sim \cos^2\theta$, producing the relational rigidity

$$R(w) \sim \frac{1}{\cos^4\theta}.$$

As rigidity diverges, the admissible transport cone collapses, and trajectories are forced to refract according to the variational law $R_1 \sin^2\theta_1 = R_2 \sin^2\theta_2$.

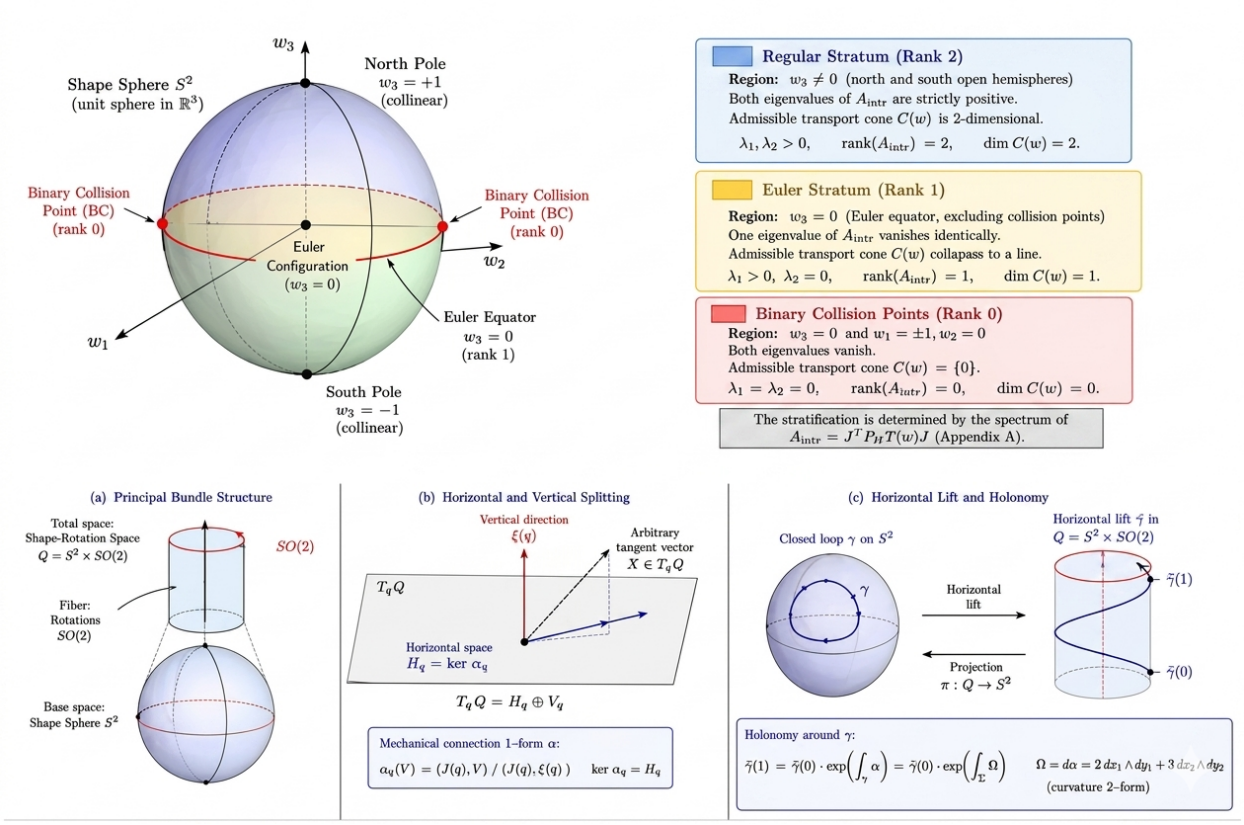


Figure 10: **Stratification of the Kendall shape sphere and admissible transport cones in the planar three-body problem.**

(a) The shape sphere S^2 with colored strata according to the rank of the intrinsic transport operator A_{intr}. In the regular stratum (blue), the admissible transport cone $C(w)$ is 2-dimensional. On the Euler equator (yellow), one eigenvalue vanishes and the cone collapses to a line (strong directional filtering). At the binary collision points (red), both eigenvalues vanish and transport is completely suppressed. The stratification is determined by the spectrum of $A_{\text{intr}} = J^T P_H T(w) J$.

(b) Geometric structure underlying the reduction: (i) principal bundle $Q = S^2 \times SO(2)$, (ii) horizontal/vertical splitting induced by the mechanical connection 1-form α, and (iii) horizontal lift of a closed loop on the shape sphere with associated holonomy.

15.6 E.6 Four-Body Realization

The same mechanism appears explicitly in the four-body problem on the stratified five-dimensional shape manifold Σ_4. On any binary collision hypersurface $\mathcal{B}_{ij}$, the intrinsic transport operator acquires the block-triangular form

$$A_{\text{intr}} \begin{pmatrix} dW_{\parallel} \\ dW_{\perp} \end{pmatrix} = \begin{pmatrix} A_{\parallel\parallel} & E_{\text{shear}} \\ 0 & 2\lambda \end{pmatrix} \begin{pmatrix} dW_{\parallel} \\ dW_{\perp} \end{pmatrix},$$

where $\lambda > 0$ is the local inertial squeeze eigenvalue. The vanishing of the lower-left block strongly suppresses the normal component while preserving admissible motion parallel to the wall. This produces continuous kinematic refraction, with transport cones flattening parallel to the hypersurface.

True Jordan-block defectiveness remains confined to lower-dimensional multi-collision strata, while the generic collinear horizon stays semi-simple.

15.7 Lattice Realization and Physical Interpretation

In the controlled non-normal lattice model, directional filtering arises at sites of large anisotropy Δ_n. These locations act as discrete rigidity interfaces, generating localized pseudospectral bulges, axis turning, and suppression of misaligned transport directions. The gluing theorem ensures that admissible bundles remain coherent across such interfaces.

Physically, directional filtering is a geometric protection mechanism. It does not rely on external forces or dissipation but emerges from the requirement that admissible motion remain closed under projection onto the allowable tangent geometry. High-rigidity regions act as natural filters that preserve only the dominant transport channels while energetically penalizing transverse excursions.

This appendix shows that directional filtering is a universal feature of the transport geometry, appearing consistently across the Grassmannian, three-body, four-body, and lattice realizations.

16 Appendix F. Transport Variational Calculus

This appendix develops the Euler–Lagrange theory for admissible transport paths on the Grassmannian.

16.1 Transport Action

Let γ be a smooth path of rank-k projectors $\Pi(s)$ on the Grassmannian. The intrinsic transport action is

$$S[\gamma] = \int_{\gamma} \sqrt{g_{\Pi}(V,V)}\, ds, \qquad V = \frac{d\Pi}{ds}.$$

Here $g_{\Pi}(V,V) = \frac{1}{2}\operatorname{Tr}(V^2)$ is the intrinsic Hilbert–Schmidt metric.

16.2 Admissible Variations

An admissible variation of γ is a smooth one-parameter family of paths $\Pi(s;\lambda)$ with $\Pi(s;0) = \Pi(s)$ and fixed endpoints. The variation vector field is

$$W(s) = \left.\frac{\partial \Pi(s;\lambda)}{\partial \lambda}\right|_{\lambda=0} \in T_{\Pi(s)}\mathrm{Gr}(k,\mathcal{H}).$$

16.3 First Variation and Euler–Lagrange Equation

The first variation of the action is

$$\delta S = \int \frac{g_\Pi(V, \nabla_s W)}{\sqrt{g_\Pi(V,V)}}\, ds,$$

where ∇_s is the covariant derivative along γ induced by the Grassmannian Levi-Civita connection.

Stationarity ($\delta S = 0$ for all admissible variations W with fixed endpoints) implies the Euler–Lagrange equation

$$\nabla_s \left(\frac{V}{\sqrt{g_\Pi(V,V)}} \right) = 0.$$

In other words, the normalized velocity $V/\sqrt{g_\Pi(V,V)}$ is parallel transported along γ with respect to the Grassmannian connection.

16.4 Interface Condition and Refraction Law

Consider a path crossing an interface between two regions with constant rigidities R_1 and R_2. Regularize the action near the interface using $R_\varepsilon = 1/(\lambda_{\min}(Q_\Pi) + \varepsilon)$. Decompose velocity into parallel and perpendicular components relative to the surviving cone.

Stationarity of the action across a thin boundary layer, together with conservation of reduced energy and continuity of curvature-carrying flux (bounded by the metric–curvature inequality, which scales with $\sin\theta_G$), yields the matching condition on the parallel components. This produces the variational refraction law

$$R_1 \sin^2\theta_1 = R_2 \sin^2\theta_2.$$

The extra sin factor (relative to ordinary Snell's law) originates directly from the curvature-carrying capacity $\propto \sin\theta_G$.

16.5 Singular Limit and Directional Protection

In the limit $R_2 \to \infty$ the refraction law forces $\theta_2 \to 0$. Transverse components are eliminated. Only the component aligned with the dominant eigenspace of I_{eff} survives. This is geometric directional filtering.

This variational calculus provides the rigorous justification for the refraction and filtering phenomena used throughout the main text.

17 Appendix G. Spectral Continuation and Persistence Theory

This appendix provides the rigorous justification for projector memory and subspace continuation using Kato perturbation theory.

17.1 Projector Continuation

Let $\Sigma(s)$ be a smooth family of self-adjoint operators on a separable Hilbert space $\mathcal{H}$ with an isolated cluster of k eigenvalues separated from the rest of the spectrum by a uniform gap $\delta > 0$.

Let $\Pi(s)$ be the associated Riesz projector onto the isolated subspace.

Theorem 17 (Projector Continuity)**.** *The map $s \mapsto \Pi(s)$ is Lipschitz continuous in the operator norm:*

$$\|\Pi(s) - \Pi(t)\| \leq C|s - t|,$$

where C depends only on the gap δ.

Proof. By the Riesz formula,

$$\Pi(s) = \frac{1}{2\pi i} \int_\Gamma (zI - \Sigma(s))^{-1}\, dz,$$

where Γ is a fixed contour in the resolvent set enclosing the isolated cluster. The resolvent is uniformly bounded on Γ by $1/\delta$. Kato perturbation theory then implies that the difference $\Pi(s) - \Pi(t)$ is controlled by the variation of the resolvent, which is Lipschitz in the perturbation parameter. □

17.2 Gap Preservation

Proposition 8 (Gap Persistence)**.** *If the spectral gap around the isolated cluster remains at least $\delta/2 > 0$ for $|s-t| < \varepsilon_0$, then the Riesz projector $\Pi(s)$ remains well-defined and the gap is preserved under small perturbations.*

This follows directly from the continuity of the spectrum under norm-continuous perturbations (Kato).

17.3 Subspace Continuity and Eigenvector Swapping

Individual eigenvectors may rotate arbitrarily or exchange identities as s varies, but the subspace they span (i.e., the range of $\Pi(s)$) varies continuously. This is the mathematical origin of **projector memory**: the subspace as a whole is more stable than any single vector.

Corollary 2. *Projector persistence (overlap $C(\Delta) = \mathrm{Tr}(\Pi(x)\Pi(x+\Delta))$) is generally stronger than single-eigenvector persistence.*

17.4 Relation to Principal Angles

The overlap is exactly quantified by the principal angles:

$$C(\Delta) = \mathrm{Tr}(\Pi_1 \Pi_2) = \sum_{i=1}^{k} \cos^2 \theta_i(\Delta).$$

Small principal angles correspond to rigid (highly persistent) channels; large angles correspond to flexible (rapidly varying) transverse directions.

17.5 Summary

Spectral continuation is governed by the persistence of isolated subspaces rather than individual eigenvectors. This justifies the use of projectors as the primary objects throughout the framework and explains why subspace coherence (projector memory) survives deformations that destroy individual mode identities.

This appendix, together with Appendices B and C, provides the complete spectral-geometric foundation for continuation principles in the main text.

18 Appendix H. Experimental Extraction Protocol

This appendix provides a fully reproducible, end-to-end protocol for extracting geometric transport quantities from raw experimental or numerical data. The protocol is deliberately divided into two distinct phases so that purely data-driven processing is cleanly separated from framework-dependent geometric interpretation.

18.1 H.1 Phase I — Experimental Extraction (Purely Data-Driven)

All operations in this phase are standard linear-algebraic and statistical procedures. No reference to the transport geometry framework is required.

18.1.1 H.1.1 Raw Data

- Input: Time series, parameter sweeps, or ensemble data (NetCDF, HDF5, or complex-valued shot records). - Record all relevant metadata (number of shots/trajectories, sampling rate, experimental parameters, noise characteristics).

18.1.2 H.1.2 Data Conditioning

- Discard corrupted or low-shot slices below a chosen threshold. - Apply standard normalization (trace normalization, covariance whitening, or ensemble averaging) as appropriate to the dataset. - Document all conditioning steps for reproducibility.

18.1.3 H.1.3 Covariance Construction

Compute the covariance matrix (or density matrix) from the conditioned data:

$$C(t) = \langle \psi(t)\psi(t)^\dagger \rangle$$

(or the equivalent $\rho(t)$).

18.1.4 H.1.4 Spectral Decomposition

Diagonalize the covariance matrix:

$$C(t) = U(t)\Lambda(t)U(t)^\dagger, \qquad \Lambda(t) = \mathrm{diag}(\lambda_1(t) \geq \cdots \geq \lambda_N(t)).$$

18.1.5 H.1.5 Projector Construction

For a chosen rank k, form the rank-k spectral projector:

$$\Pi_k(t) = U_k(t)U_k(t)^\dagger.$$

All quantities up to this point are purely spectral and can be obtained by any researcher using standard numerical linear algebra.

18.2 H.2 Phase II — Geometric Analysis (Framework-Dependent)

This phase applies the directional transport geometry framework to the projectors obtained in Phase I.

18.2.1 H.2.1 Principal-Angle Geometry

For neighboring times or parameter values, compute the principal angles via the singular values of the projector product:

$$\cos\theta_i = \sigma_i\big(\Pi_k(t)\,\Pi_k(t+\Delta t)\big),$$

or equivalently

$$\theta_i = \arccos\big(\sigma_i(\Pi_k(t)\,\Pi_k(t+\Delta t))\big).$$

18.2.2 H.2.2 Measured versus Derived Quantities

Directly measured / computed quantities (depend only on the data and linear algebra): - Covariance matrices $C(t)$ - Eigenvalues and eigenvectors - Spectral projectors $\Pi_k(t)$ - Principal angles $\theta_i(\Delta t)$ - Projector overlap / persistence $C(\Delta t) = \mathrm{Tr}(\Pi_k(t)\Pi_k(t+\Delta t))$ - Participation ratio and locked participation index

Derived geometric quantities (require the transport geometry framework): - Transport capacity $C(V) = \frac{1}{2}\,\mathrm{Tr}(V^2)$ - Relational rigidity $R = 1/(\sigma_{\min}^2 + \varepsilon)$ - Effective inertia tensor I_{eff} - Chirality $\chi(V, W)$ - Directional hierarchy $\Theta(\Delta t)$ - Curvature $K_C \sim \|\partial_t \Pi_k\|$ - Filtering strength and refraction behavior

18.2.3 H.2.3 Transport Observables

From the principal angles and projectors one obtains: - Persistence as a geometric observable - Nested directional hierarchy (rigid vs. flexible channels) - Rigidity interfaces - Refraction and directional filtering via the law $R_1 \sin^2\theta_1 = R_2 \sin^2\theta_2$ - Verification of the metric–curvature inequality

18.2.4 H.2.4 Validation

- Confirm that projector persistence exceeds single-eigenvector persistence. - Verify the metric–curvature inequality on extracted tangent variations. - Check consistency of the refraction law across identified rigidity interfaces. - Compare results across different rank choices k and regularization parameters ε.

18.3 H.3 Dependency Graph

The complete computational workflow and its logical dependencies are shown below. The horizontal line marks the transition from standard data processing to framework-dependent geometric analysis.

Raw Experimental / Simulation Data

$\downarrow$

Data Conditioning & Quality Control

$\downarrow$

Covariance (or Density) Matrix Construction

$\downarrow$

Spectral Decomposition

$\downarrow$

Rank-k Spectral Projector Π_k

Framework-Dependent Geometric Analysis

$\downarrow$

Principal Angles $\Theta(\Delta t)$

$\downarrow$

Projector Persistence $C(\Delta t) = \mathrm{Tr}(\Pi(t)\Pi(t+\Delta t))$

$\downarrow$

Transport Capacity, Rigidity, and Curvature

$\downarrow$

Directional Hierarchy and Filtering

$\downarrow$

Refraction Law and Transport Geometry

All steps above the horizontal line can be performed using only standard numerical linear algebra and require no knowledge of the transport geometry framework. All steps below the line follow directly from the Grassmannian constructions developed in Appendices B–G.

18.4 H.4 Validation Checklist

Before reporting geometric quantities, the following checks are recommended:

- Hermitian symmetry of covariance matrices
- Positive semi-definiteness of covariance matrices
- Chosen rank k and spectral gap stability
- Condition number of covariance matrices
- Bootstrap or jackknife confidence intervals on principal angles
- Stability of principal angles under small changes in rank k
- Effect of different normalization choices
- Robustness to removal of low-shot / noisy slices

- Consistency of projector persistence across multiple independent runs

18.5 H.5 Reproducibility

All results in this paper were obtained using the following computational environment and procedures:

- **Languages / Libraries**: Python 3, NumPy, SciPy, Matplotlib
- **Data format**: NetCDF / HDF5
- **Linear algebra routines**: NumPy ‘eigh‘ / ‘svd‘ (double precision)
- **Principal angle computation**: Singular values of the product $\Pi_1\Pi_2$
- **Random seeds**: Fixed for all stochastic operations (when applicable)
- **Normalization**: Trace normalization of covariance matrices unless otherwise stated
- **Regularization**: Small $\varepsilon > 0$ added to singular values when computing rigidity

Full analysis scripts and processed datasets are available upon reasonable request.

This protocol provides a transparent and reproducible bridge from raw data to the geometric transport quantities reported in the main text.

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